Online Resource 1: Supporting Tables

This PDF, `QSA_SQJ_Online_Resource_1.pdf`, accompanies *Surviving Quantum Mutants from QuCheck and Muskit Archives: Equivalent Builds, Input Limitations and Lost Histogram Information.*

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Online Resource 2 is `QSA_SQJ_Online_Resource_2.zip`; all archive paths below are relative to its `package/` root. The small cited JSON/CSV result files are shipped at their stated paths; any generated-only path is explicitly marked as regenerated by reproduce.sh at that path. The tables retain fixed-mutant-set counts and historical outcomes. The census, test-change, contract and sensitivity analyses are post hoc and exploratory. Statistic families were fixed before their own calculations, after exposure to earlier results.

S1 Reading guide and definitions

This section defines the study terms and maps each supporting analysis to its tables.

Terminology used in this resource.

Muskit's programs include conditional execution (CE) and Little Bernstein–Vazirani (LittleBV). Open Quantum Assembly Language (OpenQASM, abbreviated QASM in source signatures) describes archived circuits. Sampling uses the permuted congruential generator PCG64. Gate symbols H and X denote Hadamard and bit flip; $\mathrm{Phase}(\theta) = \mathrm{diag}(1, e^{i\theta})$ denotes a phase gate. CPU and GPU denote central and graphics processing units; KiB denotes kibibytes. A subject is one provided mutant; its source identifier locates the archived source file. Its release control is the supplied unmutated program. A slot fixes circuit arguments or state preparation; the fixed wrapper adds this preparation and differential tests to an archived program, producing a whole circuit, the build. The catalogue input is the all-zero state before that build; the application input is the state after slot preparation. A measurement basis is an output axis, Z, X or Y; the test catalogue crosses 4 slots with those axes. A test pairs a slot with a measurement basis; the test catalogue contains all tests, and a suite selects a subset. A silent subject is undetected by that entire test catalogue. A mutant set contains the subjects checked together. An input contract specifies admissible inputs and initial states of qubit registers. The executor is the circuit runner; its executor contract admits all preparable basis strings. The program-semantic contract adds documented source or property restrictions, with the executor contract where none are documented. A register is a group of qubits. Workspace qubits are selected auxiliary qubits that a workspace contract assumes start in $|0\rangle$; its mask lists them. The census classifies every subject by cause relative to its input contract. An equivalent build

(E) has full-circuit equivalence up to global phase within the unitary tolerance. An input-limited subject (I) agrees on catalogue states but differs elsewhere; it aggregates A, C and P. An admissible-input witness (A) has a state difference on an admissible basis input. A contract-limited subject (C) differs only outside the input contract; a phase-only subject (P) agrees on each admissible basis output up to its own phase, while phases differ between inputs. The existing statistics are single-qubit marginals and whole-register parity. A summary-limited subject (O) changes a catalogue state whose full histogram differs although existing statistics agree. A threshold-limited subject (T) has a resolved positive gap no larger than the threshold; D denotes detected. Infidelity above the state tolerance ϵ_s establishes a numerical state difference; it need not cause detection by existing statistics (single-qubit marginals and whole-register parity). Measurement-basis-limited (M) means different output states with identical full histograms in the catalogue measurement bases; that class is 0 at catalogue inputs here. A *behaviour group* contains subjects from the same program whose unitaries agree up to global phase in every slot within the unitary tolerance ϵ_u; groups do not establish independent failure mechanisms. QuCheck has 77 such groups and Muskit 608. Exact calculations use double-precision state vectors or unitaries and stated tolerances; they are numerical checks rather than symbolic proofs.

Notation and comparison definitions.

Let m index a subject, n its qubit count, ℓ its slot, $b = 0, 1, 2$ its Z, X or Y basis, and j an oracle statistic; test $t = 3\ell + b$. For exact subject and release-control statistic probabilities $p_j^{(m)}, p_j^{(c)}$, an exact gap is $|p_j^{(m)} - p_j^{(c)}|$; hats denote shot estimates, whose absolute difference is a stored-shot gap when computed from stored counts. The test index t is suppressed in $p_j^{(m)}, p_j^{(c)}$ and their estimates. Write $d_{m,t} = d_{m,\ell,b} = \max_j |p_j^{(m)} - p_j^{(c)}|$ and $d = \max_{\ell,b} d_{m,\ell,b}$; hats denote sampled gaps; wherever d appears in a table, it denotes a gap rather than a detection count. For subject and release-control unitaries $U_{m,\ell}, U_{c,\ell}$ and catalogue outputs $|\psi_{m,\ell}\rangle, |\psi_{c,\ell}\rangle$, define

$$u_\ell = |1 - |\operatorname{tr}(U_{m,\ell}^\dagger U_{c,\ell})|/2^n|, \qquad f_\ell = 1 - |\langle\psi_{m,\ell}|\psi_{c,\ell}\rangle|^2.$$

Here tr is the trace and † conjugate transpose. For unitary overlap deviation u_ℓ and catalogue-state infidelity f_ℓ, $u = \max_\ell u_\ell$ uses all slots, whereas $f = \max_\ell f_\ell$ and d use admissible catalogue slots; $\epsilon_u, \epsilon_s, \epsilon_o$ are their unitary, state and gap tolerances (Table S3). With $N = 8{,}192$ shots, a number of comparisons K and per-subject significance level $\alpha = 0.001$, the threshold is $\tau(K, N) = \sqrt{2\ln(4K/\alpha)/N}$; $\tau = \tau_0$ uses $K = 120$. Write q_r for qubit r and $z_r \in \{0, 1\}$ for its measured bit in the selected basis; $\oplus$ denotes exclusive OR. The pairwise parity observable is $\Pr(z_r \oplus z_s = 1)$ for $r < s$. Its historical comparison count is $L_{\text{pair}} = 12[n + 1 + \binom{n}{2}]$, with $\tau_{\text{pair}} = \tau(\max(120, L_{\text{pair}}), N)$; τ_{TV} denotes the separately adjusted total-variation threshold. For the exact subject and release-control outcome distributions $\boldsymbol{\pi}_m, \boldsymbol{\pi}_c$, a law means one such exact distribution. Total variation (TV) is $\frac{1}{2}\sum_z |\pi_m(z) - \pi_c(z)|$ over output bins z. A cell-minimum margin minimises detected cell gaps then subtracts the stated threshold; a subject-maximum margin first maximises within each subject, then minimises across subjects and subtracts that threshold. A lineage groups programs sharing source code; an algorithm family groups implementations of the same algorithm.

The historical development set supplies 60 Fourier-transform and phase-estimation subjects for selection; the evaluation set supplies 120 subjects from the other four QuCheck algorithms for reporting. Greedy selection starts at test 0, maximises newly detected development subjects, breaks ties by smallest test ID and continues through zero-gain additions until four tests. The inherited suite $T_M = [0, 11, 3, 1]$ was selected on the development set; its baseline $T_B = \{0\}$ is retained by each equal-budget comparator. Budget k counts target test invocations at fixed shots, excluding controls; an invocation runs one target test. Historical formulas use g_{base}, g_{inh} and g_{all} for baseline, inherited-suite and test-catalogue detection counts, n_{sub} for the subject denominator and $\mathcal{S}$ for a suite. The probability gap remains d, the slot index ℓ and the number of comparisons K; $\mathcal{X}$ indexes historical contrasts and v indexes programs. The full-shot arm gives each elementary test the full shot count; the fixed-budget arm shares one fixed total across the search. M and T name census classes, and J counts elementary tests. Archived data keys retain the arm codes S and B. Here $\mathcal{T}$ is the test catalogue, and $g(\mathcal{S})$ counts subjects detected by at least one test in suite $\mathcal{S}$. The uniform reference distribution is the detection-count distribution induced by equal weights on comparator suites; the uniform mean averages their detection counts, raw gain subtracts the baseline count, and excess subtracts the uniform mean from the inherited count. The inclusive tail fraction is the proportion of comparator counts at least the inherited count, including ties; it is descriptive, not a population p-value. The test-catalogue union is the union of all test detection sets; the detection matrix contains subject–test decisions. A replay repeats the same inputs, seeds and node; worker elapsed time sums measured worker execution regions, excluding end-to-end costs.

Descriptive analyses and RQ scope.

Suite selection, equal-budget contrasts, margin comparisons, C1–C4 holdout checks and source-tool outcome cross-tabs are descriptive analyses; none supplies evidence for RQ1 or RQ2. C1–C4 retain their original predicates, implementation limits and verdict codes; "incomplete" means excluded unbuildable subjects lack the planned-denominator analysis.

Resource map.

Key evidence by research question. RQ1: Tables S19 and S46–S49. RQ2: Tables S42–S44 and S50–S54. Census, budget sweep and numerical criteria: Tables S1–S3. Source comparisons, composition, execution, agreement and coverage: Tables S4–S8. Timing, comparator families, Muskit inputs and historical holdout checks: Tables S9–S17. Admissibility, behaviour groups, edit boundaries, test changes, validation, distinguishing inputs and hashes: Tables S18–S25. Duplicate signatures, single-test detections, suite times and exposure scope: Tables S26–S29. Contract comparisons, the TV diagnostic, worked suite example, reconciliation and margin nesting: Tables S30–S34. Workspace masks, original cohorts, mixed and higher-weight evidence, Pearson histogram tests, calibration and native attempts: Tables S35–S45. Slot-consistency audit and exact-law searches: Tables S46–S50. Fresh-shot detections and set-wide false-positive frequencies: Tables S51–S54. Source-tool linkage and dependencies: Table S55; native outcomes of contract-limited subjects: Table S56.

Table S1 Exact census and agreement base. E: equivalent builds at all slots; I: input-limited; O: summary-limited. Silent E/I/O/T columns partition undetected subjects; T is threshold-limited

Mutant set / program	Subjects	Detected	Silent E	Silent I	Silent O	Silent T
QuCheck	120	70	41	9	0	0
Muskit pooled	1,136	621	154	311	50	0
CE	446	233	63	100	50	0
LittleBV	333	207	44	82	0	0
QRAM	357	181	47	129	0	0

Implementation agreement / state sensitivity	QuCheck	Muskit
Agreeing subject–test cells / all cells	1,440/1,440	13,632/13,632
Minimum d above threshold	0.146447	0.066987
Detected margin	0.089933	0.010474
Undetected margin	0.056513	0.006563
Subjects with a catalogue-state difference	70/120	671/1,136
Catalogue subject–input cells with a state difference	264/480	2,531/4,544
Summary-limited subject–input cells	0	183
Positive subthreshold subject–test cells (QRAM)	—	20
Subjects contributing those cells (QRAM)	—	5

Detected gap minima here range over detected individual subject–test–statistic cells; margins subtract τ. In Table S21, the D-row gap instead first maximises over tests and statistics within each subject, then minimises over subjects. QuCheck cell/subject-max minima are 0.146447/0.250000; Muskit minima are 0.066987/0.103553. Undetected margin: τ minus the maximum undetected gap. Summary-limited subject–input cells count a subject and slot once, across all measurement bases: QuCheck has 0 of 480, and Muskit 183 of 4,544. Threshold: 0.0565. Circuits/controls are shared. Census zero tolerance: 10^{-12}; frozen Muskit agreement tolerance: 10^{-10}.

Positive subthreshold counts come from Online Resource 2, `semantic_summary.json`; all are QRAM subjects detected in another basis. Cell records: Online Resource 2, `r3/analysis/results/semantic_cells.csv`.

Table S2 Budget sweep across retained-baseline and all-suites comparator families

Scope	k	Sequence detections	Retained mean	Retained excess	Suites ≥ sequence / retained	All mean	All excess	Suites ≥ sequence / all
QuCheck	1	50	50	0	1/1	41.75	8.25	4/12
	2	63	59.36	3.64	4/11	56.97	6.03	19/66
	3	64	64.18	-0.18	37/55	63.52	0.48	131/220
	4	68	66.73	1.27	57/165	66.54	1.46	255/495
Muskit	1	529	529	0	1/1	333.25	195.75	4/12
	2	565	562.82	2.18	3/11	464.09	100.91	30/66
	3	590	580.62	9.38	18/55	529.46	60.54	94/220
	4	596	591.71	4.29	81/165	565.24	30.76	229/495
CE	1	191	191.00	0.00	1/1	118.25	72.75	3/12
	2	213	210.82	2.18	8/11	166.73	46.27	25/66
	3	222	219.51	2.49	20/55	193.46	28.54	67/220
	4	228	223.86	4.14	44/165	209.05	18.95	86/495
LittleBV	1	186	186.00	0.00	1/1	112.50	73.50	4/12
	2	186	189.82	-3.82	11/11	159.32	26.68	38/66
	3	195	193.25	1.75	27/55	180.22	14.78	136/220
	4	195	196.31	-1.31	109/165	190.62	4.38	369/495
QRAM	1	152	152.00	0.00	1/1	102.50	49.50	4/12
	2	166	162.18	3.82	3/11	138.05	27.95	30/66
	3	173	167.85	5.15	12/55	155.78	17.22	77/220
	4	173	171.54	1.46	78/165	165.57	7.43	262/495

Sequence detections use the first k tests of $[0, 11, 3, 1]$. The retained-baseline family retains test 0; the all-suites family is unconstrained. Here k counts tests. Each panel reports its uniform mean, excess over that mean, and inclusive count / all suites. Denominators: QuCheck 120 evaluation subjects (60 development subjects); Muskit 1,136, CE 446, LittleBV 333, QRAM 357. Ties and the sequence itself are included. On the separate 60 development subjects, cumulative detections by the selected 1-, 2-, 3- and 4-test sequences are 18, 38, 39, 39; these are development detections, not evaluation gains. Muskit has no development split. Test index $t = 3\ell + b$ uses Z, X and Y for $b = 0, 1, 2$. Complete distributions: Online Resource 2, `r3/analysis/results/budget_sweep.json`.

Table S3 Numerical criteria. The frozen agreement band does not round or change the strict decision $d > \tau$

Criterion	Tested quantity	Predicate
Unitary equivalence	Phase-invariant overlap deviation u_ℓ	$u_\ell \leq \epsilon_u = 10^{-9}$
State difference	Infidelity f_ℓ	$f_\ell > \epsilon_s = 10^{-10}$
Observable zero	Probability gap $d_{\ell,b}$	$d_{\ell,b} \leq \epsilon_o = 10^{-12}$
Frozen Muskit agreement	Gap d and threshold distance	$d \leq 10^{-10}$ (zero);
		$\lvert d - \tau \rvert \leq 10^{-10}$ (borderline)

A borderline cell retains its strict Boolean decision but makes agreement numerically weak under the frozen check. The state criterion and agreement band happen to use the same value; they test different quantities.

S2 Source comparisons, execution and descriptive suite checks

These tables document source composition, stored observations and the descriptive comparisons defined above.

Table S4 Closest approaches, their existing mechanisms and evaluations, and the evidence added by this finite-test-catalogue audit

Approach	Existing mechanism and evaluation	Difference studied here
QuCheck	Property generators; multi-basis circuit assertions; statistical correction	Complete fixed detection matrix, test-catalogue union and exhaustive suite distribution
Quito	Value-based coverage; specified-probability checks; detection and execution cost	Measurement basis as a test-catalogue dimension; executed release controls
MutTG	Joint suite-size and detection search; matched search evaluations	All equally sized alternatives under a baseline-preservation constraint
Fortunato et al. (2022a)	Qiskit mutation case study; equivalent mutants and oracle choice	Input-contract and statistic limits separated for supplied builds

Selected comparisons with the inspected related work; the QuCheck and MutTG rows summarise the earlier suite-selection experiment and are not evidence for RQ1 or RQ2. Cited works are listed in the article's reference list. Source inventory and inspected records: Online Resource 2, `r3/literature/RELATED_WORK.md`; bibliography: Online Resource 2, `r3/paper/refs.bib`.

Table S5 Source-derived mutant composition. Each row totals 30 subjects. Preserving is an upstream declaration; diff categories are mutually exclusive after that precedence

Family	Set	Preserving	Delete	Insert	Replace	Mixed	Total
Deutsch–Jozsa	Evaluation	10	2	8	10	0	30
Grover	Evaluation	10	1	9	10	0	30
Fourier transform	Development	10	2	13	5	0	30
Phase estimation	Development	10	1	6	12	1	30
Teleportation	Evaluation	10	1	9	10	0	30
Superdense coding	Evaluation	10	3	7	10	0	30

Table S6 Execution inventory. QuCheck records are simulator invocations, including repeated controls; target counts differ from unique subject–test cells. Holdout jobs include controls and retain build failures

QuCheck stage	Target calls	Control calls	Records
Development	720	72	792
Original evaluation, including baseline repeat	600	60	660
Added candidates	960	96	1,056
Quality-control replay	480	48	528
Sources			186
Subjects			180
Controls			6
Frozen builds			744
Unique subject–test cells			2,160
Total records			3,036
Shots per invocation			8,192
Muskit holdout quantity			**Count or time**
Planned sources (subjects + controls)			1,175 (1,172 + 3)
Sources excluded by build errors			36
Failed jobs			432
Planned and attempted jobs			14,100
Valid jobs			13,668
Valid subject jobs			13,632
Valid control jobs			36
Shots per valid job			8,192
Sampled shots			111,968,256
Planned shots			115,507,200
Archived CPU field			221.05

QuCheck's 2,040 evaluation target records sum original, added and replay calls; development is separate. Its 660 before/after comparisons include controls and repeated prior baselines. The holdout CPU value is an archived analyser accounting field; its timer region and process scope cannot be established here. It is excluded from timing comparisons. Enumeration adds no sampled jobs.

Table S7 Numerical oracle agreement on $120 \times 12 = 1{,}440$ subject–test cells

Separate observable calculation	Shot detection	Shot non-detection
Above threshold	501	0
At or below threshold	0	939

The observable calculation uses the same threshold as the shot-based differential oracle and shares frozen QASM and release controls, while implementing its own parser and numerical operations.

Table S8 Breadth counts tests detecting a subject, conditional on the 70 test-catalogue-detectable subjects. Restricted unions below count detected subjects out of 120; omitted breadth bins have 0 subjects

Breadth	2	4	5	6	7	8	9	10	11	12
Subjects	3	12	3	5	7	29	2	1	3	5

Test-catalogue restriction	Detected subjects
3-basis union at slot 0	61
3-basis union at slot 1	67
3-basis union at slot 2	66
3-basis union at slot 3	70
Z basis, all slots	62
X basis, all slots	46
Y basis, all slots	40
Z and X, all slots	70
Z and Y, all slots	70
X and Y, all slots	60

Table S9 Summed worker elapsed time over 120, 480 and 1,440 target invocations, respectively. Concurrent work is added, not converted to wall time; controls and suite construction are excluded

Suite	Tests	Detections	Total time (s)	Time/detection (s)
T_B	1	50	0.451	0.0090
T_M	4	68	1.729	0.0254
Test catalogue	12	70	4.992	0.0713

Table S10 Full forced-baseline comparison. The fixed T_M benchmark stays unchanged; for forced test a, G_a is separately reselected on development data with the original greedy rule and tie-break. Each comparator has 165 suites; $g(G_a)$ is its greedy-suite detection count

Forced test a	Mean	$\geq g(T_M)$	G_a in addition order	$g(G_a)$	$\geq g(G_a)$
0	66.73	57	$[0, 11, 3, 1]$	68	57
1	65.84	73	$[1, 9, 0, 2]$	68	73
2	65.04	57	$[2, 9, 0, 1]$	68	57
3	67.68	102	$[3, 11, 0, 1]$	68	102
4	66.28	79	$[4, 9, 0, 1]$	70	53
5	65.46	69	$[5, 9, 0, 1]$	68	69
6	68.07	108	$[6, 11, 0, 1]$	69	97
7	66.22	79	$[7, 11, 3, 0]$	68	79
8	65.20	69	$[8, 10, 0, 1]$	65	109
9	68.99	145	$[9, 1, 0, 2]$	68	145
10	66.28	79	$[10, 9, 0, 1]$	70	53
11	66.65	103	$[11, 3, 0, 1]$	68	103

Both inclusive counts include ties. The fixed T_M belongs only to comparators requiring 1 of its own tests; all G_a belong to their respective comparators. Test $t = 3\ell + b$ decodes as slot ℓ and basis b: Z=0, X=1, Y=2. Only the test-0 baseline was part of the original procedure.

Table S11 Post hoc structural suite families. Every suite has 4 tests and receives equal weight inside its listed family

Baseline constraint	Family	T_M in set?	Suites	Mean	Min	Max
Retained	Basis-balanced	Yes	96	66.97	62	70
Retained	Input-balanced	No	27	67.11	62	70
Retained	Both	No	12	67.75	65	70
Unconstrained	Basis-balanced	Yes	288	68.17	62	70
Unconstrained	Input-balanced	No	81	65.62	40	70
Unconstrained	Both	No	36	68.47	65	70

Retained means test 0 is required; unconstrained means there is no baseline-preservation constraint. Basis-balanced covers Z, X and Y; input-balanced uses distinct slots; Both meets both constraints. The fixed T_M belongs to the basis-balanced families but neither input-balanced family nor either joint family.

Table S12 Post hoc omissions of each lineage and both largest edit categories. The final columns count suites out of 165 in the fixed primary comparator; other counts refer to remaining subjects

Omitted group	Subjects left	$g(T_M)$	Union	Mean	Excess	$> g(T_M)$	$= g(T_M)$	$\geq g(T_M)$
Bell lineage	60	34	34	33.76	0.24	0	155	155
Deutsch–Jozsa lineage	90	51	53	49.97	1.03	44	16	60
Grover lineage	90	51	53	49.73	1.27	44	13	57
Declared preserving	80	68	70	66.73	1.27	44	13	57
Replacements	80	30	32	29.48	0.52	44	37	81

Each row recomputes detection counts on the remaining subjects; the suite set is unchanged.

Table S13 Muskit program/input contract from the frozen source inventory. Gate calls are static before compilation: control count, then the range across supplied sources. No body measures or resets qubits

Program	Qubits	Computation and effect of external slot preparation	Gate calls
CE	6	Conditional execution: $q_{0:2}$ control a multiqubit operation on $q_{3:5}$, followed by uncomputation. Slot preparation varies the first 2 controls; internal X, H and phase-gate preparation remains.	17; 11–18
LittleBV	6	Bernstein–Vazirani body: H/CZ/H copies the basis pattern of $q_{0:2}$ into $q_{3:5}$. Slot preparation varies only $q_{0:1}$; q_2 starts in $\|0\rangle$.	9; 8–10
QRAM	7	Controlled-swap routing between $q_{3:4}$ and $q_{5:6}$ via address q_2, then swaps into $q_{0:1}$ and applies CX/X. Slot preparation varies the latter pair; address Hadamard–phase–Hadamard gates remain.	9; 8–10

Index ranges are inclusive. Slot preparation at slot ℓ applies X wherever the little-endian integer ℓ has a set bit, before the full body; idle classical registers are retained. Inputs are ordinal, without common application semantics.

Table S14 Planned and amended complete-row denominators. Exclusions for build errors were fixed before outcomes; the original planned-denominator analysis remains incomplete. Composition uses upstream filename categories

Program	Planned	Complete	Arity	Index	Add	Remove	Replace
CE	458	446	12	0	330	12	104
LittleBV	343	333	0	10	245	9	79
QRAM	371	357	14	0	290	9	58
Total	1,172	1,136	26	10	865	30	241

Arity and Index count excluded sources by build-error type. Add/Remove/Replace counts use complete rows; no observed detection outcome determines an exclusion.

Table S15 Exposed QuCheck and outcome-blind Muskit complete-row results. $g_{\text{base}} = g(T_B)$, $g_{\text{inh}} = g(T_M)$ and $g_{\text{all}} = g(\mathcal{T})$ count detected subjects, where $g(\mathcal{S})$ counts detections by suite $\mathcal{S}$ and $\mathcal{T}$ is the test catalogue; inclusive fractions include ties and the policy

Quantity	QuCheck	Muskit
Evaluation subjects	120	1,136
$g_{\text{base}} = g(T_B)$	50	529
$g_{\text{inh}} = g(T_M)$	68	596
g_{all}, test-catalogue union	70	621
$g_{\text{inh}} - g_{\text{base}}$, gain over baseline	18	67
$g_{\text{all}} - g_{\text{inh}}$, union minus inherited	2	25
Detected-only breadth mean; minimum	7.16; 2	6.44; 1
Z-only union	62	582
Primary suites	165	165
Uniform mean μ_0	66.73	591.71
Count $\geq g_{\text{inh}}$; inclusive fraction	57; 0.3455	81; 0.4909
$g_{\text{inh}} - \mu_0$; $\mu_0 - g_{\text{base}}$	1.27; 16.73	4.29; 62.71
Cell agreements; disagreements	1,440; 0	13,632; 0
Slot-union; test-catalogue-union agreements	480; 120	4,544; 1,136
Cell-minimum margin; non-detection clearance	0.089933; 0.056513	0.010474; 0.006563

The non-detection clearance is τ minus the maximum gap among non-detected subject–test cells. Both uniform means refer to the primary baseline-preserving comparator. Breadth averages only the g_{all} detected subjects. Observable agreement is conditional on shared construction and release controls. Operator taxonomies and input semantics differ between mutant sets.

Table S16 All Muskit complete-row lineages and omissions. The primary comparator retains test 0; the unconstrained comparator drops that constraint. Each inclusive suite count includes ties and the fixed policy

Scope	n_{sub}	g_{base}	g_{inh}	g_{all}	μ_0	Primary count	μ	Unconstrained count
CE	446	191	228	233	223.86	44	209.05	86
LittleBV	333	186	195	207	196.31	109	190.62	369
QRAM	357	152	173	181	171.54	78	165.57	262
Without CE	690	338	368	388	367.85	94	356.19	298
Without LittleBV	803	343	401	414	395.40	68	374.62	166
Without QRAM	779	377	423	440	420.17	90	399.67	233

n_{sub} counts subjects; $g_{\text{base}} = g(T_B)$, $g_{\text{inh}} = g(T_M)$, $g_{\text{all}} = g(\mathcal{T})$. μ_0, μ are mean detections over baseline-preserving and unconstrained suites. The primary and unconstrained count columns count suites with $g(\mathcal{S}) \geq g_{\text{inh}}$, out of 165 and 495, respectively. Means are rounded for display; signs and verdicts use unrounded values. Every row retains the same T_B and T_M.

S3 Original Muskit holdout checks

Muskit was chosen as an available licensed mutant set with release controls, taking eligible static-register programs outside QuCheck's families. Its source rule, slot preparations, inherited suite and descriptive checks were fixed and hashed before Muskit outcomes, after QuCheck exposure. Here the historical *holdout* means that Muskit's observations had not yet been inspected when those choices were fixed; it is not fresh confirmation for the later census or test-change analyses. The buildable-subject analysis is complete; the original planned-denominator analysis is incomplete. The test catalogue detects 621/1,136 subjects, and the all-Z suite detects 582. Detection breadth, the number of detecting tests per detected subject, has mean 6.44 and minimum 1. The 2 single-test CE detections are kept in Table S27.

Table S17 states each historical predicate, its direction and its unchanged outcome. C1 asks: do decisions from stored samples agree with exact probabilities, and do subjects labelled equivalent remain undetected? C2 asks: do tests detect some but not all subjects, do several tests detect each detected subject, and does Z detect most of the detected union? C3 asks: does the mean over equal-budget suites explain at least half the inherited suite's gain, does another suite tie or beat it, and does its rank change with the comparison set? C4 asks: does a positive gain disappear against the uniform mean, or does the sign of that difference change in the specified contrasts? Its historical equivalent-label clause is retained below. The label checks are unavailable because Muskit supplies no equivalent-subject labels. The recorded C2 conjunction is partial. C4 did not replicate: the evaluated pooled and omission comparisons did not give the required nonpositive excess or sign change. The ordinal-transfer outcome is limited to particular lineages, and pooled excess does not extend to every program. These outcomes concern historical descriptive checks and supply no evidence for RQ1 or RQ2. The inclusive fractions 81/165 and 229/495 use baseline-preserving and unconstrained suites respectively, so their denominators and reference sets differ.

Reading the archived verdicts.

The verdict REPLICATION means the stated check holds; NON_REPLICATION means it fails, LINEAGE_LIMITED holds only in named lineages, PARTIAL lacks part of a conjunction, and UNAVAILABLE lacks the required labels. Read C1–C4 in order in the following table: first the predicate, then its evaluated scope, then its recorded verdict. The comparison set is the baseline-preserving suite family unless the row explicitly names the unconstrained contrast; their pooled ranks use different denominators. For C4, distinguish the within-scope sign condition from the separate lineage-dependent ordinal-transfer check; neither supplies a verdict for an unlisted scope. The notation preceding each predicate defines its comparison family and counts, and the final table retains the original directions and outcomes.

Local notation for historical predicates.

Here $g(\mathcal{S})$ counts detections by a suite; $g_a(\mathcal{S})$ restricts that count to scope a. These symbols describe historical checks only. $g_{\text{base}} = g(T_B)$, $g_{\text{inh}} = g(T_M)$, $g_{\text{all}} = g(\mathcal{T})$, n_{sub} is the subject denominator, $b_m = \sum_t \mathcal{D}(m,t)$, and $g_Z = g(\{0,3,6,9\})$.

$\mathcal{D}_{\text{shot}}$ and $\mathcal{D}_{\text{ref}}$ are shot and exact detection indicators. Here $\mathcal{I}_m(\mathcal{S}) = \mathbf{1}[\exists t \in \mathcal{S} : \mathcal{D}(m,t) = 1]$ is a subject's detection indicator, and $d_{m,t}$ is the maximum absolute probability gap for subject m and test t. $\bar{b}$ is mean breadth among detected subjects. The local subscript pool denotes the pooled result; v in the ordinal predicate indexes programs. Subscript 0 denotes the baseline-preserving comparator family; $\mathcal{C}_a$ contains all suites in the scope and comparator family indexed by a.

For each scope and family a, μ_a is the uniform mean, $\kappa_a = |\{\mathcal{S} : g_a(\mathcal{S}) \geq g_{\text{inh,a}}\}|$, and $\rho_a = \kappa_a/|\mathcal{C}_a|$, including ties and T_M.

$\mathcal{X}$ contains the reported baseline-preserving versus unconstrained and lineage-omission contrasts; the existential sensitivity rule requires a change somewhere, not in every contrast. $\mathcal{X}_{\text{C4}}$ restricts comparator swaps to pooled and each omission, and pooled-to-omission comparisons to each family; individual-program reversals are outside its evaluated scope. The comparison allowance K is distinct from the contrast sets $\mathcal{X}$ and $\mathcal{X}_{\text{C4}}$.

Interpreting the historical outcomes.

Pooled ranks $81/165 = 0.4909$ and $229/495 = 0.4626$ differ: the first uses baseline-preserving suites, the second all equal-budget suites. This change satisfies C3 without relying on mean changes. The frozen implementation compares rank and mean pairs and also reports program contrasts; the displayed rank change independently satisfies the protocol's rank criterion.

The protocol's comparator wording leaves that scope less explicit than its implementation; LittleBV's reversal is retained in Table S16 rather than used to revise the stored outcome.

The frozen C4 label clause asks whether an equivalent subject remains undetected, whereas C1 specificity requires that for every equivalent subject. Those clauses can both hold; their labels do not make them logical opposites. This fixed descriptive predicate retains its original scope; its label component is unavailable here.

The raw-gain/2 cutoff is conventional, not a practical effect threshold; a single alternative tie suffices for $\kappa_0 > 1$. Exact protocol wording and implementation: Online Resource 2, `r3/holdout/freeze/PROTOCOL.md`, sections C3 and C4; Online Resource 2, `r3/holdout/code/analyze_holdout_a1.py`. This script excludes the statically unbuildable subjects; Online Resource 2, `r3/holdout/code/analyze_holdout.py` is the base implementation listed in Table S25. Both files are supplied.

Table S17 Mathematical fixed holdout checks and unchanged archived outcomes on the buildable-subject analysis. The planned-denominator outcomes remain "incomplete"

Check	Predicate	Archived outcome
C1	$\forall m, t:\ \mathcal{D}_{\text{shot}}(m,t) = \mathcal{D}_{\text{ref}}(m,t)$, also after slot/test-catalogue OR reductions; $\nexists (m,t) : \lvert d_{m,t} - \tau \rvert \leq 10^{-10}$. Report positive and negative margins. With declared-equivalent set $E_\star$, specificity requires $\forall m \in E_\star : \mathcal{I}_m(\mathcal{T}) = 0$.	Agreement Replicated; specificity Unavailable (labels absent).
C2	$0 < g_{\text{all}} < n_{\text{sub}}$; $\bar{b} > 1 \wedge \min_{m:b_m>0} b_m \geq 2$; $g_Z > g_{\text{all}}/2$. Assess components separately; the bundle is their conjunction.	Limited union and Z majority true; redundancy false; Partial.
C3	$g_{\text{inh}} > g_{\text{base}} \wedge \mu_0 - g_{\text{base}} \geq (g_{\text{inh}} - g_{\text{base}})/2 \wedge \kappa_0 > 1$. Sensitivity requires $\exists (a,b) \in \mathcal{X} : \rho_a \neq \rho_b$; equivalently, no rank difference across those contrasts fails this component. Mean differences are reported alongside ranks.	Raw-gain pattern Replicated; sensitivity Replicated.
C4	$(g_{\text{inh}} > g_{\text{base}} \wedge g_{\text{inh}} \leq \mu_0) \ \vee\ [\exists (a,b) \in \mathcal{X}_{\text{C4}} : \text{sgn}(g_{\text{inh,a}} - \mu_a) \neq \text{sgn}(g_{\text{inh,b}} - \mu_b)] \ \vee\ [\exists m \in E_\star : \mathcal{I}_m(\mathcal{T}) = 0]$.	Not replicated in evaluated pooled and omission scopes; label component Unavailable.
Ordinal position	$g_{\text{inh,pool}} - \mu_{\text{pool},0} > 0 \wedge \forall v : g_{\text{inh},v} - \mu_{v,0} > 0$ gives consistent positive direction; positive pooled excess with a nonpositive program excess gives lineage limitation; otherwise no positive pooled direction.	Lineage limited.

S4 Contract-relative census and distinguishing-input records

The terminology key above applies to all following census tables. Muskit uses the executor contract where no program-semantic restriction is documented; superposition inputs are outside that contract. QuCheck C subjects agree within the program-semantic contract; teleportation agrees on its full admissible data subspace. Per-subject records: Online Resource 2, `r3/analysis/results/op_subjects.json`.

S4.1 Admissible and contract-violating inputs

The input audit separates admissible changes from changes that violate a program-semantic contract (Tables S18 and S24). Deutsch–Jozsa's slots cover four of the eight constant or balanced 2-bit oracles. Checking the missing four adds no detection: the silent subjects remain 10 equivalent-build and 3 contract-limited subjects.

Table S18 Application-input audit of I subjects. A: admissible-input witness; C: contract-limited; P: phase-only with no admissible-input witness. The final column counts A subjects with an existing-statistic gap $> \tau$

Program	I	A	C	P	A: $d > \tau$
QuCheck Deutsch–Jozsa	3	0	3	0	0
QuCheck Grover	3	0	3	0	0
QuCheck Teleportation	1	0	1	0	0
QuCheck Superdense coding	2	0	2	0	0
QuCheck pooled	9	0	9	0	0
Muskit CE	100	24	0	76	16
Muskit LittleBV	82	9	0	73	9
Muskit QRAM	129	97	0	32	43
Muskit pooled	311	130	0	181	68

Table S19 Subjects (behaviour groups) by program and diagnostic class: E, equivalent build; I, input-limited; O, summary-limited; T, threshold-limited; D, detected. Each of the 4 slots must match up to global phase within 10^{-9}. Groups describe build behaviour and are not independent failure mechanisms

Program	E	I	O	T	D
QuCheck Deutsch–Jozsa	10 (1)	3 (3)	0 (0)	0 (0)	17 (16)
QuCheck Grover	10 (1)	3 (3)	0 (0)	0 (0)	17 (16)
QuCheck Teleportation	10 (1)	1 (1)	0 (0)	0 (0)	19 (17)
QuCheck Superdense coding	11 (1)	2 (2)	0 (0)	0 (0)	17 (15)
QuCheck pooled	41 (4)	9 (9)	0 (0)	0 (0)	70 (64)
Muskit CE	63 (1)	100 (53)	50 (31)	0 (0)	233 (152)
Muskit LittleBV	44 (1)	82 (52)	0 (0)	0 (0)	207 (108)
Muskit QRAM	47 (1)	129 (87)	0 (0)	0 (0)	181 (122)
Muskit pooled	154 (3)	311 (192)	50 (31)	0 (0)	621 (382)

Pooled totals: QuCheck 120 (77); Muskit 1,136 (608).

Source boundaries for Table S20.

CE O source boundaries count expanded source gate calls; each row gives a boundary and the number of O subjects edited there. Boundary 18 follows the original 18 gate calls.

Gate-call boundary	O subjects
2	5
5	2
6	10
16	10
17	2
18	21

Table S20 Program × class × edit type, counting subjects. Empty classes are shown explicitly; the T residual is 0 throughout. QuCheck preserving subjects belong to the declared-preserving operator; Muskit uses Add, Remove and Replace

QuCheck					
Program	Class	Preserving	Insertion	Deletion	Replacement
Deutsch–Jozsa	E	10	0	0	0
Deutsch–Jozsa	I	0	2	0	1
Deutsch–Jozsa	O	0	0	0	0
Deutsch–Jozsa	T	0	0	0	0
Deutsch–Jozsa	D	0	6	2	9
Grover	E	10	0	0	0
Grover	I	0	3	0	0
Grover	O	0	0	0	0
Grover	T	0	0	0	0
Grover	D	0	6	1	10
Teleportation	E	10	0	0	0
Teleportation	I	0	1	0	0
Teleportation	O	0	0	0	0
Teleportation	T	0	0	0	0
Teleportation	D	0	8	1	10
Superdense coding	E	10	1	0	0
Superdense coding	I	0	1	0	1
Superdense coding	O	0	0	0	0
Superdense coding	T	0	0	0	0
Superdense coding	D	0	5	3	9

Muskit					
Program	Class	—	Add	Remove	Replace
CE	E	—	61	0	2
CE	I	—	93	1	6
CE	O	—	48	0	2
CE	T	—	0	0	0
CE	D	—	128	11	94
LittleBV	E	—	44	0	0
LittleBV	I	—	80	1	1
LittleBV	O	—	0	0	0
LittleBV	T	—	0	0	0
LittleBV	D	—	121	8	78
QRAM	E	—	46	0	1
QRAM	I	—	120	3	6
QRAM	O	—	0	0	0
QRAM	T	—	0	0	0
QRAM	D	—	124	6	51

Source lines, gate tokens, full IDs and grouping are archived in Online Resource 2, `r3/analysis/results/op_cross.csv`; Online Resource 2, `r3/analysis/results/op_subjects.csv`.

Table S21 Nearest subject maximum to each boundary: separate value and absolute-distance rows

Mutant set/class	Measure	u	f	d vs. ϵ_o	d vs. τ
QuCheck E	Value	4.441×10^{-16}	1.11×10^{-15}	3.331×10^{-16}	3.331×10^{-16}
QuCheck E	Distance	10^{-9}	10^{-10}	9.997×10^{-13}	0.056513
QuCheck I	Value	0.076120	8.882×10^{-16}	6.106×10^{-16}	6.106×10^{-16}
QuCheck I	Distance	0.076120	10^{-10}	9.994×10^{-13}	0.056513
QuCheck D	Value	0.076120	0.146447	0.250000	0.250000
QuCheck D	Distance	0.076120	0.146447	0.250000	0.193487
Muskit E	Value	6.661×10^{-16}	1.332×10^{-15}	1.332×10^{-15}	1.332×10^{-15}
Muskit E	Distance	10^{-9}	10^{-10}	9.987×10^{-13}	0.056513
Muskit I	Value	0.076120	1.554×10^{-15}	1.443×10^{-15}	1.443×10^{-15}
Muskit I	Distance	0.076120	10^{-10}	9.986×10^{-13}	0.056513
Muskit O	Value	0.076120	0.146447	7.772×10^{-16}	7.772×10^{-16}
Muskit O	Distance	0.076120	0.146447	9.992×10^{-13}	0.056513
Muskit D	Value	0.008555	0.017037	0.103553	0.103553
Muskit D	Distance	0.008555	0.017037	0.103553	0.047040

Maxima are within each subject over slots (and bases and existing statistics for d); nearest D entries minimise those subject maxima. Table S1 instead minimises detected individual cells. These are gaps; a margin subtracts its stated threshold. Empty classes have no nearest subject; T is 0 in both mutant sets. Full IDs, full precision and nearest subjects on each side: Online Resource 2, `r3/analysis/results/op_nearest.csv`. Changing ϵ_u, ϵ_s or ϵ_o individually by $\times 10$ or $/10$ changes no class and introduces 0 tolerance conflicts. The shot threshold is sensitive to an increase:

Change	QuCheck E/I/O/T/D	Muskit E/I/O/T/D
$\tau/10$	41/9/0/0/70	154/311/50/0/621
$\tau \times 10$	41/9/0/45/25	154/311/50/358/263

Online Resource 2, `r3/analysis/results/op_summary.json`.
Distance means the absolute difference between the displayed value and the stated boundary.

Table S22 Historical CE summary-limited subjects detected by pairwise parity and full-histogram total variation

Basis	Pair $> \tau$	Pair > 0	Pair $> \epsilon_o$	TV $> \tau$	TV > 0	TV $> \epsilon_o$	Stored shots
Z	16	50	16	16	50	16	16
X	42	50	42	42	50	42	42
Y	45	50	45	45	50	45	45
Union	50	50	50	50	50	50	50

Numerical examples, margin definitions and seeded reproduction: Section S4.2.

S4.2 Reproducing the seeded examples

The stored-shot result is a retrospective (q_2, q_3) reanalysis at the original unadjusted threshold. It is separate from the adjusted all-pairs and TV audit in Table S31, lies outside the fixed protocol and uses no new circuit execution. Raw positive gaps can include floating-point noise; resolved gaps exceed ϵ_o.

Kets here denote full-register basis integers. All 50 subjects have an exact pairwise-parity difference on (q_2, q_3). `000f781fb6a1c07892cb390e`: CE, slot 0, catalogue $|0\rangle$, X: $\Pr(z_2 \oplus z_3 = 1)$; exact gap 0.500000, sampled gap 0.498413 ($> \tau$); seed `1524242265652620292`. `0202259c4d869de58ca4a6aa`: LittleBV, slot 0, prebuild/application $|4\rangle$, Z: $\Pr(z_5 = 1)$; exact gap 1.000000, sampled gap 1.000000 ($> \tau$); seed `5206019462096949305`.

These are NumPy multinomial simulations with 8,192 samples per role; only one CE subject has a seeded shot example. No new backend experiment was run. Under the program-semantic contract, QuCheck has no admissible input-change example.

For stored-shot probability gaps $\hat{d}_{m,\ell,b,j}$ (subject m, slot ℓ, basis b, statistic j), the stored-shot cell-minimum margin $\min_{\hat{d}>\tau} \hat{d} - \tau$ is 0.078741 at the original τ; the subject-maximum margin $\min_m \max_{\ell,b,j} \hat{d}_{m,\ell,b,j} - \tau_{\text{pair}}$ is 0.089270 at the adjusted threshold. Their underlying gaps are 0.135254 and 0.147461. They use different cells and the same working pair, as Table S34 details; Online Resource 2, `r3/analysis/results/parity_shots.json` records each subject and input–basis cell.

Stored probability arrays are normalised by their sums without rounding or clipping. For each per-simulation label archived in `op_repairs.json`, interpret its first 8 bytes of SHA256 as a big-endian unsigned integer; initialise `np.random.default_rng(seed)`, then draw the subject multinomial followed by the independent control multinomial. Online Resource 2, `r3/analysis/results/op_repairs.json`; Online Resource 2, `r3/analysis/results/op_extended.json`.

Evidence and exposure scope.

The reported inputs and reductions contain differential measurements and suite selection. QChecker static warning-attribution supplies no outcomes to this article.

Table S23 Release controls checked against analytic expectations, followed by a separately implemented AST/numpy audit of Muskit source adaptation. Controls cover 36/36 catalogue builds. CE bits η, ζ label the basis states of q_2, q_4; QRAM v, w are the bus bits before final CX/X. The pass tolerance is 10^{-10}; state comparisons allow global phase and probability comparisons use TV

Control	Expected analytic property	Slots
Deutsch–Jozsa	Constant oracle gives 0 data; linear balanced oracle gives coefficient bits.	Matched 4/4
Grover	Marked-item success after the catalogue iteration is 1.	Matched 4/4
Teleportation	Reduced output fidelity to each catalogue data state is 1.	Matched 4/4
Superdense coding	Decoded data bits equal the catalogue classical message.	Matched 4/4
Fourier transform	Ascending-qubit circuit equals $\Pi_{\text{bit}} F \Pi_{\text{bit}}$, with Π_{bit} the bit-reversal matrix and F the positive-phase Fourier matrix.	Matched 4/4
Phase estimation	The estimation register contains the bit-reversed eigenphase integer; the eigenstate is preserved.	Matched 4/4
LittleBV	Lower secret register is preserved; upper output equals the secret.	Matched 4/4
CE	Gate-derived: basis states $z_2 = \eta$, $z_4 = \zeta$ have amplitude $e^{i\pi\zeta/4}/2$; the upper register starts at 2ζ and increments by $\eta \oplus \kappa$, $\kappa = 1 \oplus ((1 - z_0)z_1)$. The low bit flips and its neighbour is restored.	Matched 4/4
QRAM	Gate-derived: H Phase$(\pi/3)$ H gives address probabilities $3/4, 1/4$; controlled bank swaps and bus swap precede $(v, w) \mapsto (1 - v, w \oplus v)$.	Matched 4/4

In these closed forms only, $\eta, \zeta \in \{0, 1\}$ are CE bit values; QRAM v, w are the bus bits on $q_{0:1}$ immediately before the final CX/X gates. CE and QRAM expectations are closed forms derived from source gates, rather than external application specifications. Adapter passes: 4,556/4,556; builds containing `mct`: 1,652. Maximum overlap deviation: 8.882×10^{-16}; maximum phase-aligned entry error: 9.486×10^{-16}. Builds with global-phase magnitude above 10^{-10} radians: 748. Special paths cover `mct` to ancilla-free `mcx`, register reconstruction and order, static math/pi argument parsing, integer-input X slot preparations, and preserved u/cx lowering. The separately implemented interpreter executes neither source code nor the adapter parser. Online Resource 2, `r3/analysis/results/op_controls.csv`; Online Resource 2, `r3/analysis/results/op_adapters.csv`.

Table S24 Representative input-limited examples: all 9 QuCheck subjects and admissible-basis and phase-only examples from each Muskit program. A: admissible-input witness; C: contract-limited; P: phase-only with no admissible-input witness, with an outside superposition shown. Gap is the maximum existing-statistic probability difference over Z, X and Y for the chosen slot and state

Program	Full source ID	Status	Slot	Pre-build state	d
Deutsch–Jozsa	`5540ccc9cc7a8858ca45917b`	C	0	$(\|0\rangle + \|1\rangle)/\sqrt{2}$	0.353553
Deutsch–Jozsa	`663b1d13e7211d91beabea6a`	C	0	$(\|0\rangle + \|1\rangle)/\sqrt{2}$	1.000000
Deutsch–Jozsa	`953255baf4b11ac82d9d926e`	C	0	$(\|0\rangle + \|4\rangle)/\sqrt{2}$	1.000000
Grover	`25a82a398b485a86f569e50d`	C	0	$\|4\rangle$	0.353553
Grover	`a1a342888e4642d0dd70331d`	C	0	$(\|0\rangle + \|4\rangle)/\sqrt{2}$	0.176777
Grover	`b1bfc15ed0e95b11faf7d48e`	C	0	$\|4\rangle$	1.000000
Teleportation	`c96ba53fb2d4a472293ae210`	C	0	$(\|0\rangle + \|2\rangle)/\sqrt{2}$	0.500000
Superdense coding	`34b06e6d4d982d18ca0b5812`	C	0	$(\|0\rangle + \|3\rangle)/\sqrt{2}$	1.000000
Superdense coding	`f22155415ba0e18c492cd073`	C	0	$(\|0\rangle + \|1\rangle)/\sqrt{2}$	0.353553
CE	`0fb76edbfd2109a521a8a3c5`	A	0	$\|33\rangle$	0.707107
CE	`063eec1e4b0e55fc02b06a85`	P	0	$(\|0\rangle + \|16\rangle)/\sqrt{2}$	1.000000
LittleBV	`0202259c4d869de58ca4a6aa`	A	0	$\|4\rangle$	1.000000
LittleBV	`02c2b95d3f7709f6a53b5139`	P	0	$(\|0\rangle + \|8\rangle)/\sqrt{2}$	0.353553
QRAM	`071d396bc01aa5c444ab5d2f`	A	0	$\|40\rangle$	1.000000
QRAM	`161b81a154c721aa20034709`	P	0	$(\|0\rangle + \|28\rangle)/\sqrt{2}$	0.433013

Integer kets use the complete register. For Muskit, application input is the pre-build integer XOR slot; the displayed A examples all use slot 0. C and P examples are outside the established input contract. The full I inventory comprises QuCheck A/C/P = 0/9/0 and Muskit A/C/P = 130/0/181. Each stored I row retains the full ID, chosen slot, distinguishing input, gap and admissible-input audit: Online Resource 2, `r3/analysis/results/op_subjects.csv`; Online Resource 2, `r3/analysis/results/op_subjects.json`.

Table S25 Historical original identifiers; public copies may differ. Original hashes do not certify translated or summarised copies or external preregistration. Online Resource 2, `PAPER_SOURCE_MAP.json` maps original to distribution hashes and identifies translations, summaries and withheld originals

`r3/holdout/freeze/PROTOCOL.md`	
Original:	`552f606f486bb59b1c6a35f850b543ad9bd43bf187d6ac45d2b7cdcc75d9bc94`
Distribution:	`db279ca660f17cd6fb4ea5d3999c66e01254d077e975f9a4da3cff1d30da3818`
`r3/holdout/freeze_h2/CODE.sha256`	
Original:	`10dbb13df2cdd9e607b9a4d8b6465a2364b1fd731cbeea82f4e49d42730b0120`
Distribution:	`cf8fa279f083a3522263b14930fac6fd5871ff6cf92643511207a39fd0a6d7fc`
`r3/holdout/amendments/A1_STILLBORN.md`	
Original:	`1364d5d07382c3274bdb634f0cfd6af7b8d5247ecc1201d83eb0845aa47c05c3`
Distribution:	`36eaf90b0fa3ea380336293ad0e30f4da31a5ad1c1655d5847934a83cf7f6b13`
`r3/holdout/amendments/A1b_NOTE.md`	
Original:	`1c60d086ae372161e2986801142d05fc866c013320ee94e6e38b064596a6bf99`
Distribution:	`765983de0e2aaf8c4b62d7c338a9c97706e6c331d2c2d84f908b65aeff336e72`
`r3/holdout/amendments/A1.sha256`	
Original:	`40f0c491c93be7d2262c4551c9a898a44fa08ee8bcc02b6bb3772d0583e81863`
`r3/holdout/code/adapter.py`	
Original:	`ce92386bf85b2dca94bd2afe5166299924ea072d8f2157db54985e4c3ce71b01`
`r3/holdout/code/run_holdout.py`	
Original:	`9dda23e5aa4462db987ca7bc62cbe1bad7b42817d1c1e4525aa6e8546731c7f7`
`r3/holdout/code/analyze_holdout.py`	
Original:	`c5dc7bd1f5725116de67df266d065370f37ae045f1b09c8be7fda6f19144da4b`
Distribution:	`02d2367b745f854f1d4944efaae7de27474e964279d8c1e8c1cbc2bc725a3119`
`r3/holdout/tests/test_holdout.py`	
Original:	`d488c063eafd490b33bf94d7d1b4a1250299af289299e2a2bd42246991bb4b00`
Distribution:	`602c916cd5da479c93cb6f524b3181cded99c63f26309ea5b104a87c169a15a1`
`r3/holdout/env/lock.txt`	
Original:	`e4e53b873e00e3f17e8f0048798a8644a3b20410a1e326e79d7ea19fc8dfc80d`
`r3/holdout/env/ENV.json`	
Original:	`2bce5142432f13397bda7b9f9ed1922889d52d9ad40f2e80f42e1ab906d1a6b3`
Distribution:	`9f0122eebefcb750a1bb64a196407dfc6272cd3105c8a93a3d9395c1681ad283`
`r3/holdout/env/requirements-hashed.txt`	
Original:	`b9d6e4695251f5280d48167f56a14e9da24fc87357546f9de9fe0403d1d44cc4`
`r3/holdout/env/dry-inventory.json`	
Original:	`eaf13ad50ebdebb77a2b5c9ac87cf3d522d1c37ffcdfd840422c29c0d48c4049`
`r3/holdout/env/dry-inventory-summary.json`	
Original:	`ae62683460f95802d26ef50dd87845df4b0934d70ab8bada4e15ae149f3a820e`
`r3/holdout/env/tests.log`	
Original:	`aa15318d27ad0c1f1f9cdccfee652bf865fe0d391884a7e329fd893bef2450c2`
Distribution:	`0ef31f8aadeab8cbac22ec53bc85ab6f48311416a62d8bd02f2645f9a5915a16`
`r3/holdout/docs/H2_REVIEW.md`	
Original:	`81b8bde65aab944eb4973f94b488475458680de29a33f96e880726b59bec3545`
Distribution:	`1261dd1e67d38be68c502532494485cac0a888e5feb7260ff5f5d6f581636309`
`r3/holdout/tests/fixtures/bell.py`	
Original:	`d3de6f4c36924ea3367ad9bfceac283b85e33d83a724b8cbb0fb3b83692dc7a8`
`r3/holdout/tests/fixtures/gates.py`	
Original:	`fe769e6d371d0d78ef582bc1614b710f18356c839e64cf92b0931eb2ab2a19ec`
`r3/holdout/tests/fixtures/ghz.py`	
Original:	`3834b6a73a96f6f14d7da11e561176371b39abaddb1d43e22cda11a268529844`
`r3/holdout/tests/fixtures/hst.py`	
Original:	`ac54d63ec85b27da5cf49e929a4d67a65f9e7c9bb8f61601fd7e2d0fc3d74bc0`
`r3/holdout/tests/fixtures/mct.py`	
Original:	`cd9e4147c5759f36f45550eab649dea640b6fc043db20c601fa460604c8ecaaa`

Original: historical original digest; Distribution: public distribution digest where different. Protocol and amendments preceded Muskit outcomes; historical full manifest: Online Resource 2, `r3/holdout/freeze_h2/CODE.sha256`. Analysis code and QASM source digests: Online Resource 2, `r3/analysis/results/op_summary.json`.

Table S26 All repeated QuCheck QASM signatures. Each signature joins family and ordered raw build digests; this differs from unitary behaviour grouping

1b12c262695016f4604db1962d9ada069e3f634b8515f122b9984d53d637efd0 b32f01a113a924a010321e78, eabd9c589e892233385cf5da
7013523df3a266ac38f199356897cd7a639c85bf69e00423259fb05ac245e735 1b536f03b36cffcfd1ffd620, 8261d973aef9451dd136cceb
76f80e2592936bdaa02d7585f4bf00063651c8699e62d30728c887c44644841b 5fd35e1068cadb4eac60892e, 70e8c573b8bcd6b9be24c3bd

Complete membership and source identifiers: Online Resource 2, `r3/analysis/results/non_witness_summary.json`; collapsed suite outcomes: Online Resource 2, `r3/analysis/results/non_witness_summary.json`. Collapsed baseline/inherited suite/test-catalogue union: 49, 67, 68; mean 65.24, tail fraction 0.3091 (Online Resource 2, `r3/paper_numbers/generated_core_tables.md`, T10).

Table S27 Both 1-test CE detections. Columns are test IDs in order; these rows defeat the all-detected-subject redundancy condition

Source ID	0	1	2	3	4	5	6	7	8	9	10	11
5c0a0b9c1546f2e25cfe40bd	0	0	0	0	0	0	0	0	0	1	0	0
3aeccae34edefa0bffee8946	0	0	0	0	0	0	1	0	0	0	0	0

5c0a0b9c1546f2e25cfe40bd: ExperimentalData/CE/Mutants/19AddGate_cswap_inGap_1_.py
3aeccae34edefa0bffee8946: ExperimentalData/CE/Mutants/227AddGate_ccx_inGap_12_.py
Complete rows: Online Resource 2, `r3/holdout/results/ANALYSIS_A1.json`.

Table S28 Muskit suite-time and census-cost scopes. Suite sums reuse 1 execution; census costs come from 1 later local analysis

Recorded quantity	Value
Baseline-preserving suites	165
Target calls per suite	4,544
Target time (s): minimum / median / maximum	20.368 / 21.619 / 22.199
Inherited suite time (s) / ascending rank	21.460 / 49
Maximum/minimum spread	9.0%
Census wall time (s) / peak RSS (KiB)	22.21 / 401,912
Full analysis wall time (s) / peak RSS (KiB)	26.22 / 584,648
Builds / maximum qubits	5,024 / 7

Suite time sums successful subject `CELL` records' `backend.time_taken`; controls, starts, failures, construction, transport and scheduling are excluded. Every suite's full sum and detections: Online Resource 2, `r3/analysis/results/budget_sweep.json`, `muskit_target_time.suite_results`. Census includes input search, CE extension, adapter checks and groups; full analysis adds analytic controls and output writing, excluding the final summary write, selftests and independent review. Peak RSS uses Linux `resource.getrusage` in KiB. Python `3.14.4`, NumPy `2.5.0`, `OPENBLAS_NUM_THREADS=1`; environment and scope: Online Resource 2, `r3/analysis/results/op_summary.json`, `cost`. Dense matrices use 16×4^n bytes each; caches and simultaneous slots increase peak memory beyond 1 matrix.

Table S29 Evidence and exposure. Chronology and complete protocol/code/amendment digests are retained in the supporting hash inventory

Evidence layer	Fixed information, exposure and scope
QuCheck execution	Development determined the inherited suite. Comparator fixed after its evaluation raw gain, before added tests; original GPU binary/interpreter patch unpinned.
Record reduction	3,036 stored histograms rebuilt matrices and suite counts; same-node, same-seed record replay compares stored records, without fresh draws.
Muskit execution	Source rule, slot preparations, sequence, checks, code and static exclusions fixed before Muskit outcomes, after QuCheck exposure; pinned CPU environment.
Numerical recomputation	Separate parser/gates on preserved builds; 36/36 analytic control checks and 4,556/4,556 source-to-unitary adapter checks.
Retrospective census	Unitary/state/input/observable census, detections, groups and budget sweep followed both outcomes; seeded test-change shots are seeded NumPy simulations.

QuCheck: Python `3.11`, Qiskit `1.2.4`, GPU Aer `0.15.1`; Muskit: Python `3.11.15`, same Qiskit/Aer, CPU statevector. Optimisation level 0 and recorded seeds are fixed; manifests distinguish reduction, recomputation and original simulator execution.

S5 Exploratory test changes and contract sensitivity

The full-histogram TV and all-pairs parity comparisons use separate adjusted thresholds. TV at the original threshold detects equivalent builds; applying the large TV threshold to existing statistics also loses some detections. The split-alpha Pearson histogram test below is the principal histogram comparison. The TV comparison records a separate diagnostic with its own design limits.

Table S30 Historical statistic families fixed before their calculations, after exposure to earlier CE findings

Program	Qubits	Added pairs	L_{pair}	τ_{pair}	τ_{TV}
CE	6	180	264	0.058191	0.115294
LittleBV	6	180	264	0.058191	0.115294
QRAM	7	252	348	0.058768	0.155316
Deutsch–Jozsa	3	36	84	0.056513	0.061821
Grover	3	36	84	0.056513	0.061821
Teleportation	3	36	84	0.056513	0.061821
Superdense coding	2	12	48	0.056513	0.056513

Each family covers 12 tests per subject. All pairs are fixed before this calculation, after exposure to earlier CE findings. The thresholds control each stated family per subject, not both arms or the complete mutant sets. Definition and event counts: Online Resource 2, `r3/analysis/results/rev_observables.json`.

Table S31 Historical conservative TV comparison, retained with its arm-design limitations

Mutant set	Mode	Subjects	Original	Pairs	TV orig.	TV adj.	E FP orig.	D lost
QuCheck	Exact	120	70	70	70	70	0	0
QuCheck	Shots	120	70	70	70	70	0	0
Muskit	Exact	1,136	621	671	671	667	0	4
Muskit	Shots	1,136	621	671	904	668	65	3

Pairs means existing statistics plus all pairwise parity observables at the adjusted threshold. TV orig. uses the original threshold; TV adj. applies the histogram-adjusted threshold to that arm. Both adjusted arms detect 0 E subjects and no additional silent E/I subjects. E FP orig. is false detections among equivalent builds at the original TV threshold; D lost is original detections lost at the adjusted TV threshold. All O subjects are detected by both adjusted arms. These stored control histograms are reused and do not estimate an independent false-positive rate. The historical TV arm applied its large threshold to existing statistics as well as TV; its QRAM losses are an arm-design artefact. The split-alpha Pearson histogram test in Table S43 retains all original detections; independent control draws are calibrated in Table S44. Full class, program and control counts: Online Resource 2, `r3/analysis/results/rev_observables.json`.

S6 Historical suite example and outcome records

The suite T_M is inherited from the exposed QuCheck development set; its raw gain is the increase over baseline detections, while excess is the difference from the uniform mean over equal-budget comparators. Its QuCheck raw gain decomposes as $18 = 16.73 + 1.27$; the full budget sweep describes these suites without evaluating a transferable selection method.

S6.1 Worked example: a detected subject missed by the inherited suite

The given `superdense_coding_m9.py` deletes `qc.z(0)` from message encoding, so the input-to-measurement path has a concrete missing operation. At slot 3, message 11, execution prepares a Bell pair, encodes the message, decodes, rotates into the selected basis, and samples both qubits. The classical message bits are γ_1, γ_2. The release control decodes to both bits 1, whereas the subject loses γ_1 on q_0, so Z at test 9 detects the subject while Y at test 11 does not. Table S32 gives the sampled probabilities. The vector is $(\Pr(z_0 = 1), \Pr(z_1 = 1), \Pr(\text{odd parity}))$; $\hat{d}$ is the largest absolute difference between corresponding subject and release-control components.

Table S32 Changing the measurement basis detects the illustrative subject whose encoding gate is deleted

Slot / basis	Sampled subject probabilities	Sampled control probabilities	$\hat{d}$	Detects
3 / Z	$(0.0000, 1.0000, 1.0000)$	$(1.0000, 1.0000, 0.0000)$	1.000000	1
3 / Y	$(0.5056, 0.5009, 0.5057)$	$(0.5055, 0.4968, 0.5031)$	0.004028	0

The exact Z differences are $(1, 0, 1)$, while all exact Y differences are 0, matching the shot decisions against $\tau \approx 0.0565$. The fixed $T_M = \{0, 11, 3, 1\}$ misses this subject; replacing its final X test with the slot-3 Z test gives $\{0, 11, 3, 9\}$. That equal-budget alternative detects 70 subjects, including both Bell subjects missed by T_M, though only this deletion is explained here. This example was chosen after evaluation; it does not explain or validate the earlier development-set choices.

Further numerical census examples.

LittleBV `Add102.py`, source ID `0202259c4d869de58ca4a6aa`, inserts $\mathrm{CZ}(q_2, q_5)$ and is compared with release control `19e8137ff9361a09de206867`. Across slots, $u = 0.500000$, $f = 0.000000$, $d = 2.22 \times 10^{-16}$: it is non-E but I. At slot 0, application input $|4\rangle$ yields infidelity 1 and Z gap $d = 1.000000$ on q_5, so it is A under both contracts. A seeded multinomial example gives $\hat{d} = 1.000000$; the full-shot arm detects it in 1,000/1,000 repetitions under either contract. Under the Muskit executor contract, the fixed-budget arm cannot reject by construction; its archived zero events are an implementation check. Under the slot-consistent contract, the fixed-budget arm detects this subject in 1,000/1,000 repetitions. For contrast, CE source

`000f781fb6a1c07892cb390e`, relative to control `0727e4d7dd62e11e9f4ff109`, has $u = 0.292893$, $f = 0.500000$, $d = 2.22 \times 10^{-16}$ and is O under both contracts. Its catalogue X histogram at slot 0 changes, and the Pearson histogram test detects it; (q_2, q_3) explains the correlation, with exact gap 0.500000 and seeded multinomial gap 0.498413 (Table S22).

Source-tool outcome cross-tab.

`FAILED` is a literal outcome in Muskit's original test report, including `FAILED WITH P-Value nan`; the latter records an undefined p-value, not a valid rejection. Exact member and SHA`256` linkage gives 1,129 source subjects, with 7 unlinked subjects omitted without near-name substitution. Under executor, the linked E, C and O classes contain $154 + 0 + 50 = 204$ subjects, or 18.1%; all subjects in those classes are linked to such a label. The source-tool label and the wrapper class describe different tests and input domains; their agreement in total is not a source-tool false-positive rate or a revised mutation score. The slot-consistent equivalent-build, contract-limited and summary-limited classes contain 154, 171 and 50 subjects, respectively. Of these, 374 are linked (33.1%), and one is unlinked. The unlinked equivalent-build, contract-limited or summary-limited subject is excluded from the source cross-tab by the exact member/SHA`256` rule above; this exclusion does not change the census. QuCheck's full source-tool ratio is not derivable without its missing outcome-to-instance crosswalk (Table S55).

S7 Exploratory census and measurement analyses

The following analyses address contract sensitivity, histogram testing, set-wide false-positive frequencies and native QuCheck property-based tests after earlier outcomes were known. The preserved observations and protocols document exposure; the historical holdout does not independently confirm these exploratory analyses.

Cell minima and subject maxima.

The stored-shot cell minimum is 0.135254. Its margin is 0.078741 at $\tau = 0.056513$ and 0.077063 at the adjusted pair threshold $\tau = 0.058191$. The minimum subject maximum is 0.147461. Its original-threshold margin is 0.090948 and its adjusted-threshold margin is 0.089270. The exact minimum subject maximum is 0.146447, with adjusted margin 0.088255.

The cell-minimum example uses slot 3, basis Y and pair (q_2, q_3); the subject-maximum example is a different subject, using slot 1, basis X and the same pair. The former subject has a larger gap, 0.155396, at slot 0 in Y. Thus the differing reported minima use different subjects and aggregation orders. For every one of the 50 O subjects, this pair attains the maximum over all pairs and existing statistics. The other 14 pairs have stored-shot maximum 0.029175, below the original threshold.

Table S33 Reconciliation of different aggregations, denominators and units retained in the historical tables

Quantity	Values	Interpretation
QuCheck detected gap	0.146447 / 0.250000	Individual-cell minimum / minimum of within-subject maxima; subtract the stated threshold only when reporting a margin.
Muskit detected gap	0.066987 / 0.103553	Individual-cell minimum / minimum of within-subject maxima; subtract the stated threshold only when reporting a margin.
Inclusive fractions	81/165 vs. 229/495	Baseline-preserving and unconstrained suites have different denominators.
DJ existing builds	124	Each of 30 subjects and 1 release control has 4 existing oracle builds; added oracles are a separate check.
Comparison allowance	117 / 120	$(12 + 1)(8 + 1)$ scalar comparisons; rounded upward conservatively.
Input-limited subset / whole set	311 / 1,136	The subset contains I subjects; the whole-set census checks every subject and admissible slot.
A and cumulative test changes	130: 68/76/118	A establishes a state difference. Input / input plus pairs / input plus histogram need separate statistic checks.
Test-change scopes	50 / 8	Catalogue-input O detection and extra A detection after input search are different populations.
CE E units	63 / 1	Subjects / behaviour groups; the second count is not an independent-mechanism count.
CE O source boundaries	50 total	Table S20 prints gate-call boundaries and subject counts as separate columns.
Development sequence	18, 38, 39, 39	Cumulative detections on 60 development subjects for the selected 1-, 2-, 3-, 4-test sequences.
Gain decomposition	18 = 16.73 + 1.27	Raw gain above baseline = uniform mean gain above baseline + inherited-suite excess above that mean.

Each value retains its stated aggregation and denominator. Pair gap and margin nesting is detailed in Tables S22 and S34. Full precision and selectors: Online Resource 2, `r3/analysis/results/rev2_reader.json`.

Table S34 Gap and margin are different quantities; cell minima and minima of subject maxima use different nesting

Mode	Aggregation	Threshold	Gap	τ	Margin
Shots	Cell minimum	Original	0.135254	0.056513	0.078741
Shots	Cell minimum	Adjusted	0.135254	0.058191	0.077063
Shots	Subject maximum	Original	0.147461	0.056513	0.090948
Shots	Subject maximum	Adjusted	0.147461	0.058191	0.089270
Exact	Cell minimum	Original	0.146447	0.056513	0.089933
Exact	Cell minimum	Adjusted	0.146447	0.058191	0.088255
Exact	Subject maximum	Original	0.146447	0.056513	0.089933
Exact	Subject maximum	Adjusted	0.146447	0.058191	0.088255

The cell-minimum margin takes the smallest detected cell gap and subtracts its threshold; the subject-maximum margin first takes each detected subject's maximum gap, then the smallest of those maxima, and subtracts its threshold. The pair (q_2, q_3) attains every subject maximum; existing statistics and the other pairs do not increase it. Explicit formulas appear in Table S22. Online Resource 2, `r3/analysis/results/rev2_margins_subjects.csv`.

Slot-inconsistent sensitivities.

CE lower fixes $q_{0:2}$ and QRAM lower fixes $q_{0:1}$. Both are explicitly slot-inconsistent: each excludes slots 1, 2 and 3, removing 9 catalogue tests per subject by construction. CE loses 4,014 subject–test cells and moves 14 D subjects into 3 A and 11 C; QRAM loses 3,213 cells and moves 19 D into 12 A and 7 C. These are excluded preparations, not newly discovered survival mechanisms. All per-program masks, admissible and removed slots and exact transition IDs are in Online Resource 2, `r3/analysis/results/rev3_audit_masks.csv` and Online Resource 2, `r3/analysis/results/rev3_audit_D_transition_ids.csv`. Combined mask labels list CE, LittleBV and QRAM in that order. Write $q_{r:s}$ for all qubits from r through s inclusively; a fixed-zero mask prepares each in $|0\rangle$. An asterisk marks a slot-inconsistent mask.

S7.1 Workspace masks and admissibility

The following shared definitions apply to Tables S35–S39 and S46–S50. Upper and lower identify the higher or lower qubit-index ranges fixed to zero, as specified below; joint labels list CE, LittleBV and QRAM in that order. An asterisk marks excluded slots. Each mask is a sensitivity assumption rather than a recovered application specification. A single-program restriction leaves the other programs under the executor contract.

Whole-set workspace sensitivity.

Table S35 reclassifies all 1,136 Muskit subjects under 9 contracts: the executor contract; separate CE-lower, CE-upper, LittleBV-upper, QRAM-lower and QRAM-upper restrictions; and the joint CE-upper/LittleBV-upper/QRAM-lower, CE-upper/LittleBV-upper/QRAM-upper and CE-lower/LittleBV-upper/QRAM-lower

restrictions. CE-lower fixes $q_{0:2}$ in $|0\rangle$; CE-upper and LittleBV-upper fix $q_{3:5}$ in $|0\rangle$. QRAM-lower fixes $q_{0:1}$ in $|0\rangle$; QRAM-upper fixes $q_{3:6}$ in $|0\rangle$.

For a fixed-qubit mask W, a slot ℓ is admissible only if $\ell \,\&\, W = 0$. An admissible application basis string ξ also has zeros on those qubits, and its pre-build preparation is $\xi \oplus \ell$. Undoing a slot preparation does not make an excluded slot admissible. E has highest priority and remains fixed globally; D and O are reconsidered using admissible slots at the catalogue input, with the original threshold for D. T denotes a positive but subthreshold gap. Otherwise, search for A over all admissible basis inputs, then P over their coherent span, and assign C if neither state difference exists. Every contract has 0 T subjects.

Under the executor contract the counts are E=154, A=130, C=0, P=181, O=50 and D=621. Restrictions make 33 formerly D subjects silent: 14 in CE-lower and 19 in QRAM-lower. They comprise 15 new A subjects and 18 new C subjects. They remain silent at catalogue inputs; input search detects the new A subjects. The full contract view includes these subjects, rather than reusing only the original silent cohort.

Table S39 separates each contract's own A population from the fixed original A cohort. For the executor contract, input search, addition of pairs, and cumulative addition of Pearson histogram tests give exact-law detections for 68, 76 and 118 of 130 subjects. The joint CE-upper/LittleBV-upper/QRAM-lower contract instead gives 62, 62 and 103 of its own 113 A subjects. With QRAM-upper in that combination, all 8 of 8 A subjects have exact-law detections at every stage. Changing denominators is not evidence of improved detection. These exact-distribution diagnostics do not control multiplicity over searched inputs or over the entire subject set. The exact-law column of main-text Table 4 applies the following deterministic rules of the generating code (`analysis/code/rev3_contracts.py`, `arms()`), all at catalogue scale with $N =$ 8,192: input rung, maximum gap of the existing statistics above τ_0; + pairs rung, maximum gap including pairwise parities above $\tau(K, N)$ with $K = \max\{120, 12[n + 1 + \binom{n}{2}]\}$; + histogram rung, maximum scalar gap above $\tau_{\text{split}}(K, N)$ at the same K, or the Pearson test on exact pseudo-counts with $p_{\text{val}} < \alpha/(2 \times 12)$. The search-wide per-subject adjustment applies only to the two fresh-shot columns.

Table S35 Whole-set Muskit census under all 9 input contracts; every row contains 1,136 subjects

Input contract	E	A	C	P	O	D
Executor	154	130	0	181	50	621
CE lower*	154	131	79	118	47	607
CE upper	154	108	32	171	50	621
LittleBV upper	154	126	31	154	50	621
QRAM lower*	154	139	19	172	50	602
QRAM upper	154	34	108	169	50	621
Upper/upper/lower*	154	113	82	135	50	602
Slot-consistent	154	8	171	132	50	621
Lower/upper/lower*	154	136	129	82	47	588

Mask definitions and admissibility: Section S7.1. E retains its full-circuit meaning. T and the catalogue measurement-basis-limited class are 0 in every row. CE lower and QRAM lower move 14 and 19 formerly D subjects into silent classes. Source: Online Resource 2, `r3/analysis/results/rev2_census.json`.

Table S36 Complete CE rows for the same 9-contract census

Input contract	E	A	C	P	O	D
Executor	63	24	0	76	50	233
CE lower*	63	25	79	13	47	219
CE upper	63	2	32	66	50	233
LittleBV upper	63	24	0	76	50	233
QRAM lower*	63	24	0	76	50	233
QRAM upper	63	24	0	76	50	233
Upper/upper/lower*	63	2	32	66	50	233
Slot-consistent	63	2	32	66	50	233
Lower/upper/lower*	63	25	79	13	47	219

Mask definitions and admissibility: Section S7.1. E retains its full-circuit meaning. Rows repeat the executor row when a restriction affects only other programs. Source: Online Resource 2, `r3/analysis/results/rev2_census_tables.csv`.

Table S37 Complete LittleBV rows for the same 9-contract census

Input contract	E	A	C	P	O	D
Executor	44	9	0	73	0	207
CE lower*	44	9	0	73	0	207
CE upper	44	9	0	73	0	207
LittleBV upper	44	5	31	46	0	207
QRAM lower*	44	9	0	73	0	207
QRAM upper	44	9	0	73	0	207
Upper/upper/lower*	44	5	31	46	0	207
Slot-consistent	44	5	31	46	0	207
Lower/upper/lower*	44	5	31	46	0	207

Mask definitions and admissibility: Section S7.1. E retains its full-circuit meaning. Rows repeat the executor row when a restriction affects only other programs. Source: Online Resource 2, `r3/analysis/results/rev2_census_tables.csv`.

Table S38 Complete QRAM rows for the same 9-contract census

Input contract	E	A	C	P	O	D
Executor	47	97	0	32	0	181
CE lower*	47	97	0	32	0	181
CE upper	47	97	0	32	0	181
LittleBV upper	47	97	0	32	0	181
QRAM lower*	47	106	19	23	0	162
QRAM upper	47	1	108	20	0	181
Upper/upper/lower*	47	106	19	23	0	162
Slot-consistent	47	1	108	20	0	181
Lower/upper/lower*	47	106	19	23	0	162

Mask definitions and admissibility: Section S7.1. E retains its full-circuit meaning. Rows repeat the executor row when a restriction affects only other programs. Source: Online Resource 2, `r3/analysis/results/rev2_census_tables.csv`.

Table S39 All contract-specific A detections versus the unchanged original A cohort (130 subjects)

Contract	Own A	input	+ pairs	+ histogram	Original A eligible	input	+ pairs	+ histogram
Executor	130	68	76	118	130	68	76	118
CE lower*	131	69	77	119	127	66	73	115
CE upper	108	54	54	96	108	54	54	96
LittleBV upper	126	64	72	114	126	64	72	114
QRAM lower*	139	80	88	129	127	68	76	117
QRAM upper	34	26	34	34	34	26	34	34
Upper/upper/lower*	113	62	62	103	101	50	50	91
Slot-consistent	8	8	8	8	8	8	8	8
Lower/upper/lower*	136	77	85	126	120	62	69	110

Left: each contract's newly classified A subjects. Right: fixed executor A cohort; eligible counts only members with an admissible basis-state difference. Each last column combines existing statistics, pairs and the Pearson histogram test under split-alpha allocation; omitting pairs gives the same totals. Mask definitions and admissibility: Section S7.1. No new D occurs within the original A cohort. New silent subjects from executor D explain why own-A totals can increase under restrictions. Input searches are exploratory. Online Resource 2, `r3/analysis/results/rev2_census_repair_tables.csv`.

Mixed local measurement settings.

A Pauli string is a tensor product of $\mathbb{I}$, X, Y and Z; $\mathbb{I}$ is the identity and weight counts non-identity factors. A homogeneous setting uses the same axis on all qubits; a mixed local measurement setting may use different axes.

All product settings in $\{Z, X, Y\}^n$ give exactly $3n + 3^2\binom{n}{2}$ distinct Pauli observables of weight at most two when only marginals and pairwise parity observables are retained. Summing projectors on unused qubits gives the identity, so their local axes cannot change these statistics. The corresponding probability gap is half the absolute Pauli-expectation gap.

Table S40 covers every original silent non-E subject: 370 in total, comprising 9 QuCheck subjects under the program-semantic contract and 361 Muskit subjects under the executor contract. Catalogue inputs and searches over admissible basis inputs are reported separately. At catalogue inputs, mixed statistics detect the O subjects but do not detect the input-limited subjects. This is an exact numerical search, with threshold comparisons, rather than a new mixed-basis shot experiment.

For QRAM input searches, mixed pairs raise pair-only detections from 40 to 43, but add 0 subjects to the executor-wide union of 76 detections from existing statistics and homogeneous pairs. The remaining 12 subjects have state differences at other inputs but identical Pauli statistics of weight at most two. Changing only the local pair bases cannot detect them. P subjects instead agree up to global phase on every admissible basis input.

A separate post hoc, exploratory calculation computed all 16,384 Pauli coefficients per fixed input without stopping at its first gap and found a mixed correlation of weight four for every residual subject, with parity-probability gap 0.866025. This calculation followed exposure to the negative result for weight at most two. The minimum nonzero weight is conditional on each chosen input; it is not a minimum over

all inputs. These inputs are admissible under the executor contract but not necessarily under every workspace assumption. Measurement-basis-limited (M) means that different output states have identical full histograms in the catalogue measurement bases. The catalogue measurement-basis-limited count is 0; the residual QRAM subjects receive this additional classification only after input search. The catalogue O subjects remain summary-limited because their homogeneous full histograms already differ.

Table S40 Original silent non-E cohort: all local-Pauli marginal and pair statistics, with catalogue inputs separated from input search

Mutant set / program	Class	Subjects	Catalogue	Input search	Homogeneous TV $> \epsilon_o$
QuCheck	C	9	0	0	0
Muskit CE	A	24	0	24	24
Muskit LittleBV	A	9	0	9	9
Muskit QRAM	A	97	0	43	85
Muskit	P	181	0	0	0
Muskit CE	O	50	50	50	50
QRAM residual subset	A	12	0	0	0

Mixed columns use all Pauli strings of weight at most two and the adjusted mixed-family threshold; nonzero, original-threshold and homogeneous-pair-adjusted counts agree here. The residual row is a subset of QRAM A, not an extra group. This fixed original cohort contains 361 Muskit and 9 QuCheck subjects. Separate phase-only input sensitivity uses single-qubit $|+\rangle$ inputs and the all-$|+\rangle$ input before each slot, outside the basis-input contract: it detects 181 of 181 phase-only subjects with existing statistics (Online Resource 2, `r3/analysis/results/rev_phase.json`). Newly silent subjects under restrictions are included in Table S41. Online Resource 2, `r3/analysis/results/rev2_mixed_subjects.csv`.

Table S41 Complete silent non-E Muskit populations under all contracts: mixed local-Pauli diagnostics of weight at most two

Input contract	Silent non-E	Catalogue	Input search
Executor	361	50	126
CE lower*	375	47	124
CE upper	361	50	104
LittleBV upper	361	50	122
QRAM lower*	380	50	138
QRAM upper	361	50	84
Upper/upper/lower*	380	50	112
Slot-consistent	361	50	58
Lower/upper/lower*	394	47	132

Mask labels follow Section S7.1. Denominators come from each new census, including the 33 newly silent executor-D subjects (15 A, 18 C). The full view contains 6,668 subject–contract–scope rows, including both input scopes. Restrictions create no new catalogue mixed detection; the silent O subjects remain summary-limited. Online Resource 2, `r3/analysis/results/rev2_mixed_contracts.csv`.

Table S42 All 12 QRAM residuals have weight-four mixed correlations at the fixed inputs shown

Subject ID prefix	Application input	Pauli string	Min. weight	Parity gap
`235d4fbe`	54	`XXIIXIY`	4	0.866025
`3e8a5fd9`	28	`XXIIXIY`	4	0.866025
`85a92824`	28	`XXIIXIY`	4	0.866025
`85cf7c69`	52	`XXIIXIY`	4	0.866025
`a725fc13`	112	`IXIIXXY`	4	0.866025
`bdd29863`	9	`IXIIXXY`	4	0.866025
`c58fe0f4`	92	`IXIIXXY`	4	0.866025
`cbd4670b`	8	`IXIIXXY`	4	0.866025
`cc8024de`	88	`IXIIXXY`	4	0.866025
`e04a08bb`	28	`XXIIXIY`	4	0.866025
`e5d0af69`	92	`IXIIXXY`	4	0.866025
`f91ab35a`	8	`IXIIXXY`	4	0.866025

Every row uses slot 0 and the executor contract. Pauli strings run from q_6 to q_0, with I the identity. The local setting is `XXZZXZY` for `XXIIXIY` and `ZXZZXXY` for `IXIIXXY`; identity qubits are completed with Z. The weight minimum is at that fixed input, not across every input. All marginal and pair gaps and homogeneous histogram gaps remain numerically 0 over the admissible basis-input search. The all-Pauli expansion is post hoc and exploratory, with exposure to the negative result for weight at most two. Full IDs and cross-checks of tensors and rotations: Online Resource 2, `r3/analysis/results/rev2_higherweight.csv`.

S7.2 Histogram comparison and error control

The comparison count and error allocation distinguish catalogue decisions from the exploratory histogram tests. The historical allocation uses $(12 + 1)(8 + 1) = 117$ scalar comparisons and rounds upward to $K = 120$ (Table S33). The factors include the catalogue tests plus the replay allocation, and single-qubit marginals plus whole-register parity. For the supplied builds, the conservative catalogue count also covers $12(n + 1) \leq 120$.

Split-alpha Pearson histogram test.

For observed subject and release-control integer counts a_z, c_z in output bin z from independent samples of equal size $N = 8{,}192$, use the actual two-sample Pearson statistic

$$\mathcal{B} = \{z : a_z + c_z > 0\}, \qquad Q = \sum_{z \in \mathcal{B}} \frac{(a_z - c_z)^2}{a_z + c_z},$$

$$\nu = |\mathcal{B}| - 1, \qquad p_{\mathrm{val}} = \Pr(\chi^2_\nu \geq Q).$$

Only pooled-zero bins are removed; bins are not merged. If $\nu = 0$, set $p_{\mathrm{val}} = 1$. With $\alpha = 0.001$, allocate $\alpha/2$ to existing statistics and $\alpha/2$ to the 12 histogram comparisons,

rejecting a histogram only when $p_{\mathrm{val}} < \alpha/(2 \times 12)$. The existing-statistic threshold is

$$\tau_{\mathrm{split}}(K, N) = \sqrt{\frac{2\ln(4K/(\alpha/2))}{N}}, \qquad K = \max\{120, 12(n+1)\}.$$

Here K is the number of comparisons. All supplied builds have $n \leq 9$, so $12(n+1) \leq 120$ and this maximum equals the catalogue $K = 120$. Detection is the union of a strict gap exceedance and a histogram rejection. In the cumulative test-change comparison including pairs, replace the allowance by $\max\{120, 12[n + 1 + \binom{n}{2}]\}$ and retain the same α split. No histogram total-variation threshold is applied to a marginal.

The exact-distribution arm uses deterministic pseudo-counts $N\pi_{m,z}$ and $N\pi_{c,z}$ after removing numerical roundoff zeros and renormalising. It is a diagnostic, not an observed hypothesis test or a power estimate. Pearson histogram test tails are asymptotic: cells with an expected count $(a_z + c_z)/2$ below 5 occur in 64/15,072 exact cells and 899/15,072 stored-shot cells. Sparse bins therefore limit finite-sample error guarantees.

Table S43 gives identical subject decisions for the exact and stored-shot arms: detections rise from 691 to 741 among 1,256, adding the 50 O subjects and losing 0 existing detections. Observed E detections are 0/195. The Pearson histogram test is the principal histogram comparison; the conservative total-variation comparison records a separate diagnostic.

Table S43 Split-alpha Pearson histogram test: exact pseudo-count and stored-shot subject decisions agree

Mutant set	Class	Subjects	Existing	Split existing	Pearson	Union
QuCheck	E	41	0	0	0	0
QuCheck	C	9	0	0	0	0
QuCheck	D	70	70	70	70	70
Muskit	E	154	0	0	0	0
Muskit	A	130	0	0	0	0
Muskit	P	181	0	0	0	0
Muskit	O	50	0	0	50	50
Muskit	D	621	621	621	621	621

Only the O class gains detections; no existing detection is lost, and stored-shot E detections remain 0/195. Exact pseudo-counts $N\boldsymbol{\pi}_m, N\boldsymbol{\pi}_c$ are diagnostic, not observations or power estimates. Sparse expected-bin cells (expected count below 5): exact 64/15,072, stored shots 899/15,072. The Pearson histogram test uses an asymptotic approximation without pooling sparse bins. Online Resource 2, `r3/analysis/results/rev2_histogram_cells.csv`.

Independent-draw multinomial calibration.

Table S44 uses 1,000 repetitions for each of 195 E subjects and every test, totalling 2,340,000 subject–control sample pairs. Each pair contains independent draws $\boldsymbol{a} \sim$ Multinomial(8,192, $\boldsymbol{\pi}_m$) and $\boldsymbol{c} \sim$ Multinomial(8,192, $\boldsymbol{\pi}_c$) from the respective archived

exact laws. Draws are independent across subjects, test cells and repetitions; controls are never reused. Alternative comparisons use the same draw pairs. Streams use seed 20,261,101, stable subject digests and separate cell and role streams with NumPy SeedSequence and `PCG64`; each repetition index spans the entire subject set.

For the Pearson histogram test alone and for the split-alpha union, per-subject false-positive counts range from 0 to 3 per 1,000 repetitions. There are 75 events among 195,000 subject repetitions, involving 62 subjects. The set-wide false-positive frequency counts repetitions with any false positive: 71/1,000, or 0.071000, with Wilson 95% interval $[0.056670, 0.088614]$. α was not divided across the subjects. Per-subject error control at α is not simultaneous control across the subject set; sparse-bin approximation also remains relevant.

The existing-statistic, adjusted-pair and adjusted-total-variation comparisons each have 0/1,000 set-wide events; a zero-event one-sided exact 95% upper bound is 0.00299125, not proof of zero error probability. The unadjusted total-variation diagnostic has 1,000/1,000 set-wide events. Individual subject intervals are not simultaneous intervals. All rates condition on archived noiseless laws and this subject set, rather than hardware or unseen programs.

Original draws used NumPy 2.3.5; verification fixed the archived NumPy 1.26.4 environment and checked stored draw checksums and events. Multinomial replay can differ across NumPy versions even with the same seed, so reproducing these events requires the recorded implementation and version.

Table S44 Set-wide false-positive events in 1,000 independent repetitions

Statistic family	QuCheck	Muskit	All subjects	Wilson low	Wilson high
Existing statistics	0	0	0	0.000000	0.003827
Existing + pairs, adjusted	0	0	0	0.000000	0.003827
Existing + TV, adjusted	0	0	0	0.000000	0.003827
Existing + TV, original threshold	0	1,000	1,000	0.996173	1.000000
Split-alpha Pearson union	13	58	71	0.056670	0.088614

Each repetition includes all 195 E subjects and 12 tests per subject; the denominator in every count column is 1,000. A set-wide event means at least one E subject is detected. Zero-event arms have one-sided exact 95% upper bound 0.00299125. Per-subject error control does not add a correction across all 195 subjects. Intervals condition on the archived noiseless exact laws. Online Resource 2, `r3/analysis/results/rev2_calibration_summary.json`.

Native QuCheck PBT.

Native QuCheck property-based testing (PBT) preserves the archived generators, preconditions and assertions, separately from the fixed wrapper. Its exploratory protocol was written and hashed before execution: 9 properties, 4 accepted inputs per property, seed 20,261,101, $N = 8{,}192$ per measurement and native $\alpha = 0.01$. Native Fisher comparisons, Holm–Bonferroni adjustment and most-frequent checks were retained. The first attempt disabled native optimisation and produced 93 `KeyError` failures

when duplicate circuits lost measurement identifiers. Exceptions were not counted as detections or native non-detections.

A repair protocol was separately written and hashed before the corrected run. It restored the native default `run_optimization=True` for circuit deduplication and measurement mapping, retaining native transpilation optimisation level 0. Source code, properties, generators, inputs, preconditions, seed and statistical rules were unchanged. These are failed and corrected attempts within the same study, not independent repetitions. The first attempt retains count archives for 31 completed cases and error logs for the 93 failures; its wrapper did not save the failures' raw counts, leaving that evidence gap.

Table S45 reports only the corrected run: 68/120 detections, agreement with the test catalogue for 116/120, E detections 0/41, release-control detections 0/4, and 0 execution errors. Agreement compares recorded native property-failure outcomes with the test-catalogue union; it does not independently validate native significance calculations.

Native generator size parameters span Deutsch–Jozsa 3–10 qubits and Grover 4–7 qubits, rather than being restricted to the fixed wrapper's dimensions. The E, C and D classes describe the fixed builds and are not equivalence claims across these generator ranges. Including controls, the corrected run used 3,255 measurement circuits and 26,664,960 shots, with 12–48 circuits per subject. Only shots per measurement match; total budgets are unmatched. Although corrected count archives are retained, their dumps lack the assertion-to-circuit mapping needed to recompute native p-values independently.

The failed attempt logged 4,464 measurement circuits and 36,569,088 shots, separately from the corrected budget above. Every algorithm family has 17/30 native detections. The offline CPU environment used Python 3.11.15, Qiskit 1.2.4, Aer 0.15.1, NumPy 1.26.4 and SciPy 1.17.1; all 296 archived Python source files compared byte-identically.

Base protocol SHA256: `182bf1f841d9fde4db012e9da9e13f85 9ad3dff93049b5fbf669ab8ff2786622`.

Repair protocol SHA256: `c54470f8f4f74e40d8833fcb260c82c8 ff02d5996596273c1b94661fdb1b0faf`.

Table S45 Corrected native QuCheck PBT run, grouped by the fixed-wrapper census class

Class / population	Subjects	Detected	Native non-detections	Error
E	41	0	41	0
C	9	1	8	0
D	70	67	3	0
All subjects	120	68	52	0
Release controls	4	0	4	0

Native PBT detects 68/120 subjects and agrees with the test catalogue on 116/120. Native-only: Grover m8; test-catalogue-only: Grover m11, teleportation m9 and m13. The E, C and D classes classify fixed wrapper builds and do not claim equivalence across native generator sizes. Total circuit budgets differ. Online Resource 2, `r3/analysis/results/rev2_native.json`.

Separately implemented checks and preserved evidence.

A separate raw-QASM parser, gate engine, count decoder and gamma-tail calculation checked 30,144 histogram cells and 15,252 distinct archived histograms; multinomial verification checked all 4,680 cell-role streams and their event arrays. The evidence-preservation self-check covers 13,689 protected original files and all 496 archived numerical values. These checks support the stored observations and arithmetic; native assertion p-values remain outside the independently reconstructed evidence because the stored dumps lack their assertion-to-circuit map. Reproduction requires the archived sampling environment, as described in Online Resource 2, `r3/analysis/REV2_ENVIRONMENT.md`; source observations and analysis outputs remain separate from rendered tables.

S8 Slot-consistent contracts and search-wide fresh-shot evidence

The executor contract admits all preparable basis inputs. Defined after exposure within the audited masks, the slot-consistent contract fixes CE and LittleBV $q_{3:5}$ and QRAM $q_{3:6}$ just after slot preparation, before the program body, preserving every slot and test. It fixes no qubit varied by a slot or documented preparation at that boundary; the masks are assumptions, not recovered semantic specifications. Body-internal coherent gates are distinct from external preparation. Coherent phase probes can violate even these masks and remain outside both main contracts.

Table S46 Full Muskit contract sensitivity: subjects (behaviour groups)

Mask	E	A	C	P	O	D
Executor	154 (3)	130 (95)	0 (0)	181 (97)	50 (31)	621 (382)
CE lower*	154 (3)	131 (96)	79 (37)	118 (73)	47 (29)	607 (370)
CE upper	154 (3)	108 (76)	32 (24)	171 (92)	50 (31)	621 (382)
LittleBV upper	154 (3)	126 (91)	31 (20)	154 (81)	50 (31)	621 (382)
QRAM lower*	154 (3)	139 (101)	19 (18)	172 (89)	50 (31)	602 (366)
QRAM upper	154 (3)	34 (30)	108 (72)	169 (90)	50 (31)	621 (382)
Upper/upper/lower*	154 (3)	113 (78)	82 (62)	135 (68)	50 (31)	602 (366)
Slot-consistent	154 (3)	8 (7)	171 (116)	132 (69)	50 (31)	621 (382)
Lower/upper/lower*	154 (3)	136 (98)	129 (75)	82 (49)	47 (29)	588 (354)

All rows retain the original 608 groups; T and M are 0. Mask definitions and slot consistency: Section S7.1. Per-program and subject rows: Online Resource 2, `r3/analysis/results/rev3_contract_census.csv` and Online Resource 2, `r3/analysis/results/rev3_contract_subjects.csv`.

Table S47 Complete CE contract census: subjects (groups)

Mask	E	A	C	P	O	D
Executor	63 (1)	24 (21)	0 (0)	76 (32)	50 (31)	233 (152)
CE lower*	63 (1)	25 (22)	79 (37)	13 (8)	47 (29)	219 (140)
CE upper	63 (1)	2 (2)	32 (24)	66 (27)	50 (31)	233 (152)
LittleBV upper	63 (1)	24 (21)	0 (0)	76 (32)	50 (31)	233 (152)
QRAM lower*	63 (1)	24 (21)	0 (0)	76 (32)	50 (31)	233 (152)
QRAM upper	63 (1)	24 (21)	0 (0)	76 (32)	50 (31)	233 (152)
Upper/upper/lower*	63 (1)	2 (2)	32 (24)	66 (27)	50 (31)	233 (152)
Slot-consistent	63 (1)	2 (2)	32 (24)	66 (27)	50 (31)	233 (152)
Lower/upper/lower*	63 (1)	25 (22)	79 (37)	13 (8)	47 (29)	219 (140)

The same groups are retained under every mask. Mask definitions and slot consistency: Section S7.1. Full source IDs: Online Resource 2, `r3/analysis/results/rev3_contract_subjects.csv`.

Table S48 Complete LittleBV contract census: subjects (groups)

Mask	E	A	C	P	O	D
Executor	44 (1)	9 (8)	0 (0)	73 (44)	0 (0)	207 (108)
CE lower*	44 (1)	9 (8)	0 (0)	73 (44)	0 (0)	207 (108)
CE upper	44 (1)	9 (8)	0 (0)	73 (44)	0 (0)	207 (108)
LittleBV upper	44 (1)	5 (4)	31 (20)	46 (28)	0 (0)	207 (108)
QRAM lower*	44 (1)	9 (8)	0 (0)	73 (44)	0 (0)	207 (108)
QRAM upper	44 (1)	9 (8)	0 (0)	73 (44)	0 (0)	207 (108)
Upper/upper/lower*	44 (1)	5 (4)	31 (20)	46 (28)	0 (0)	207 (108)
Slot-consistent	44 (1)	5 (4)	31 (20)	46 (28)	0 (0)	207 (108)
Lower/upper/lower*	44 (1)	5 (4)	31 (20)	46 (28)	0 (0)	207 (108)

The same groups are retained under every mask. Mask definitions and slot consistency: Section S7.1. Full source IDs: Online Resource 2, `r3/analysis/results/rev3_contract_subjects.csv`.

Table S49 Complete QRAM contract census: subjects (groups)

Mask	E	A	C	P	O	D
Executor	47 (1)	97 (66)	0 (0)	32 (21)	0 (0)	181 (122)
CE lower*	47 (1)	97 (66)	0 (0)	32 (21)	0 (0)	181 (122)
CE upper	47 (1)	97 (66)	0 (0)	32 (21)	0 (0)	181 (122)
LittleBV upper	47 (1)	97 (66)	0 (0)	32 (21)	0 (0)	181 (122)
QRAM lower*	47 (1)	106 (72)	19 (18)	23 (13)	0 (0)	162 (106)
QRAM upper	47 (1)	1 (1)	108 (72)	20 (14)	0 (0)	181 (122)
Upper/upper/lower*	47 (1)	106 (72)	19 (18)	23 (13)	0 (0)	162 (106)
Slot-consistent	47 (1)	1 (1)	108 (72)	20 (14)	0 (0)	181 (122)
Lower/upper/lower*	47 (1)	106 (72)	19 (18)	23 (13)	0 (0)	162 (106)

The same groups are retained under every mask. Mask definitions and slot consistency: Section S7.1. Full source IDs: Online Resource 2, `r3/analysis/results/rev3_contract_subjects.csv`.

Table S50 Exact-law detection of A subjects for all Muskit contracts: subjects (groups)

Mask	Own A	input	+ pairs	+ histogram
Executor	130 (95)	68 (58)	76 (65)	118 (89)
CE lower*	131 (96)	69 (59)	77 (66)	119 (90)
CE upper	108 (76)	54 (46)	54 (46)	96 (70)
LittleBV upper	126 (91)	64 (54)	72 (61)	114 (85)
QRAM lower*	139 (101)	80 (67)	88 (74)	129 (97)
QRAM upper	34 (30)	26 (23)	34 (30)	34 (30)
Upper/upper/lower*	113 (78)	62 (51)	62 (51)	103 (74)
Slot-consistent	8 (7)	8 (7)	8 (7)	8 (7)
Lower/upper/lower*	136 (98)	77 (64)	85 (71)	126 (94)

These are deterministic numerical detections; masks follow Section S7.1. The exact-law column of main-text Table 4 applies the following deterministic rules of the generating code (`analysis/code/rev3_contracts.py`, `arms()`), all at catalogue scale with $N = 8{,}192$: input rung, maximum gap of the existing statistics above τ_0; + pairs rung, maximum gap including pairwise parities above $\tau(K, N)$ with $K = \max\{120, 12[n + 1 + \binom{n}{2}]\}$; + histogram rung, maximum scalar gap above $\tau_{\text{split}}(K, N)$ at the same K, or the Pearson test on exact pseudo-counts with $p_{\text{val}} < \alpha/(2 \times 12)$. The search-wide per-subject adjustment applies only to the two fresh-shot columns. Every scenario and cohort, including original A and newly silent subjects: Online Resource 2, `r3/analysis/results/rev3_audit_all_ladder_groups.csv`.

S8.1 Fresh-shot allocation and scopes

A role means subject or release control, and a rung is a cumulative family in the nested search. Recorded PCG64 seeds provide independent streams for each subject, elementary test, role and contract pairing. QuCheck retains its executor domain in both Muskit contract pairings; their recorded contract indices select separate streams. The full-shot arm gives each elementary test 8,192 samples per role. The fixed-budget arm divides the test-catalogue budget $12 \times 8{,}192 = 98{,}304$ per role (196,608 including control) over the same full search, using floor allocation and a one-shot remainder in fixed order. The input rung uses existing statistics, the + pairs rung adds every pairwise parity and the + histogram rung adds full-histogram tests. For N_{hist} histograms and total circuit qubit count n, $L = N_{\mathrm{hist}}[n+1+\binom{n}{2}]$ counts scalar tests and $J = L + N_{\mathrm{hist}}$ counts elementary tests; the number of scalar comparisons is $K = L$. An elementary test is one scalar statistic or one histogram at an input, slot and basis. Scalar tests use single-qubit marginals, whole-register parity and pairwise parities. The input, + pairs and + histogram rungs share one search-wide per-subject $\alpha = 0.001$; half covers all scalar comparisons and half all histograms, without reallocation or renewed α at earlier rungs. Scalar samples are independent binomial pushforwards of full multinomial laws; histogram tests use independent full-law multinomial draws. Every searched test is sampled, without stopping or sharing controls. For CE and LittleBV under the executor contract $J = 17{,}664$, $N = 5$–6 in the fixed-budget arm; QRAM $J = 46{,}080$, $N = 2$–3. Slot-consistent searches use $J = 2{,}208$, $N = 44$–45 and $J = 2{,}880$, $N = 34$–35 respectively. The supplied protocol is Online Resource 2, `r3/rev3_evidence/PROTOCOL.md`, SHA256 `2aa69018f25f1395997d0695248ae874e9ec7270ba54b78a03c24037b30ce65e`. Under the Muskit executor contract, the fixed-budget arm has $N \leq 6$, so $\tau_{\mathrm{split}}(K, N) > 1$ and $Q \leq 2N \leq 2 \times 6$ cannot attain $p_{\mathrm{val}} < \alpha/(2N_{\mathrm{hist}})$, as derived in the manuscript. Its zero detections are an analytical design consequence; the archived simulation is retained only as an implementation check. All elementary tests have separate draws; sharing bit strings across statistics at one circuit setting would give a different allocation, which was not evaluated. Search coordinates use direct prebuild columns for QuCheck and the exclusive OR of the application input and slot for Muskit. These simulations add no circuit/backend execution. Each detection frequency uses 1,000 repetitions. Each group is represented by its member with the smallest source identifier; detection of any other member does not count for that group. Group counts are descriptive, not independent mechanisms. The Pearson histogram test removes pooled-zero bins without sparse-bin pooling; its asymptotic p-values do not guarantee exact finite-sample α. The search-wide adjustment covers one subject's whole search, not all 195 E subjects together. Tables S51–S53 share these definitions. Counts at detection frequencies of at least 0.95 and 0.5 agree. Subject intervals and pointwise Wilson and Clopper–Pearson intervals for every scope are supplied in Online Resource 2, `r3/analysis/results/rev3_ladder_summary.json` and Online Resource 2, `r3/analysis/results/rev3_ladder_subjects.csv`; neither interval family provides simultaneous coverage. Tables give the arm and cumulative rung in separate columns.

Table S51 Fresh-shot detections and set-wide false-positive frequencies under the executor contract for Muskit and QuCheck; counts are subjects (behaviour groups)

Muskit contract	Scope	Arm	Rung	A detections	Mean A detection frequency	Set-wide false-positive frequency
Executor	Muskit	Full-shot	input	68 (58)	0.523077	0.000 (0.000)
Executor	Muskit	Full-shot	+ pairs	76 (65)	0.584615	0.000 (0.000)
Executor	Muskit	Full-shot	+ histogram	118 (89)	0.907731	0.052 (0.001)
Executor	Muskit	Fixed-budget	input	0 (0)	0.000000	0.000 (0.000)
Executor	Muskit	Fixed-budget	+ pairs	0 (0)	0.000000	0.000 (0.000)
Executor	Muskit	Fixed-budget	+ histogram	0 (0)	0.000000	0.000 (0.000)
Executor	QuCheck	Full-shot	input	2 (2)	1.000000	0.000 (0.000)
Executor	QuCheck	Full-shot	+ pairs	2 (2)	1.000000	0.000 (0.000)
Executor	QuCheck	Full-shot	+ histogram	2 (2)	1.000000	0.019 (0.004)
Executor	QuCheck	Fixed-budget	input	1 (1)	0.579000	0.000 (0.000)
Executor	QuCheck	Fixed-budget	+ pairs	1 (1)	0.579000	0.000 (0.000)
Executor	QuCheck	Fixed-budget	+ histogram	2 (2)	1.000000	0.013 (0.002)

Contract labels name the Muskit contract paired with QuCheck's unchanged executor domain. Shared rate definitions and interval scope: Section S8.1.

Table S52 Fresh-shot detections and set-wide false-positive frequencies for the pooled executor scope and the slot-consistent Muskit scope; counts are subjects (behaviour groups)

Muskit contract	Scope	Arm	Rung	A detections	Mean A detection frequency	Set-wide false-positive frequency
Executor	Pooled	Full-shot	input	70 (60)	0.530303	0.000 (0.000)
Executor	Pooled	Full-shot	+ pairs	78 (67)	0.590909	0.000 (0.000)
Executor	Pooled	Full-shot	+ histogram	120 (91)	0.909129	0.071 (0.005)
Executor	Pooled	Fixed-budget	input	1 (1)	0.008773	0.000 (0.000)
Executor	Pooled	Fixed-budget	+ pairs	1 (1)	0.008773	0.000 (0.000)
Executor	Pooled	Fixed-budget	+ histogram	2 (2)	0.015152	0.013 (0.002)
Slot-consistent	Muskit	Full-shot	input	8 (7)	1.000000	0.000 (0.000)
Slot-consistent	Muskit	Full-shot	+ pairs	8 (7)	1.000000	0.000 (0.000)
Slot-consistent	Muskit	Full-shot	+ histogram	8 (7)	1.000000	0.059 (0.001)
Slot-consistent	Muskit	Fixed-budget	input	6 (5)	0.750250	0.000 (0.000)
Slot-consistent	Muskit	Fixed-budget	+ pairs	8 (7)	1.000000	0.000 (0.000)
Slot-consistent	Muskit	Fixed-budget	+ histogram	8 (7)	1.000000	0.002 (0.000)

Contract labels name the Muskit contract paired with QuCheck's unchanged executor domain. Shared rate definitions and interval scope: Section S8.1.

Table S53 Fresh-shot detections and set-wide false-positive frequencies for QuCheck and pooled scopes paired with the slot-consistent Muskit contract; counts are subjects (behaviour groups)

Muskit contract	Scope	Arm	Rung	A detections	Mean A detection frequency	Set-wide false-positive frequency
Slot-consistent	QuCheck	Full-shot	input	2 (2)	1.000000	0.000 (0.000)
Slot-consistent	QuCheck	Full-shot	+ pairs	2 (2)	1.000000	0.000 (0.000)
Slot-consistent	QuCheck	Full-shot	+ histogram	2 (2)	1.000000	0.010 (0.001)
Slot-consistent	QuCheck	Fixed-budget	input	1 (1)	0.568000	0.000 (0.000)
Slot-consistent	QuCheck	Fixed-budget	+ pairs	1 (1)	0.568000	0.000 (0.000)
Slot-consistent	QuCheck	Fixed-budget	+ histogram	2 (2)	1.000000	0.014 (0.001)
Slot-consistent	Pooled	Full-shot	input	10 (9)	1.000000	0.000 (0.000)
Slot-consistent	Pooled	Full-shot	+ pairs	10 (9)	1.000000	0.000 (0.000)
Slot-consistent	Pooled	Full-shot	+ histogram	10 (9)	1.000000	0.069 (0.002)
Slot-consistent	Pooled	Fixed-budget	input	7 (6)	0.713800	0.000 (0.000)
Slot-consistent	Pooled	Fixed-budget	+ pairs	9 (8)	0.913600	0.000 (0.000)
Slot-consistent	Pooled	Fixed-budget	+ histogram	10 (9)	1.000000	0.016 (0.001)

Contract labels name the Muskit contract paired with QuCheck's unchanged executor domain. Shared rate definitions and interval scope: Section S8.1.

Table S54 Histogram-rung false-positive frequencies pooled over QuCheck and Muskit under both Muskit contracts: subject and behaviour-group means, set-wide frequencies and Wilson intervals

Muskit contract	Arm	Unit	E units	Mean detection frequency	Set-wide false-positive frequency	Wilson interval
Executor	Full-shot	subjects	195	0.00037949	0.071	[0.056670, 0.088614]
Executor	Full-shot	groups	7	0.00071429	0.005	[0.002138, 0.011651]
Executor	Fixed-budget	subjects	195	0.00006667	0.013	[0.007613, 0.022114]
Executor	Fixed-budget	groups	7	0.00028571	0.002	[0.000549, 0.007263]
Slot-consistent	Full-shot	subjects	195	0.00036923	0.069	[0.054884, 0.086415]
Slot-consistent	Full-shot	groups	7	0.00028571	0.002	[0.000549, 0.007263]
Slot-consistent	Fixed-budget	subjects	195	0.00008205	0.016	[0.009872, 0.025832]
Slot-consistent	Fixed-budget	groups	7	0.00014286	0.001	[0.000177, 0.005643]

Contract labels name the Muskit contract paired with QuCheck's unchanged executor domain. Both earlier rungs have 0 mean and 0 set-wide false-positive events in every scope. Every rung has 0 E units at detection-frequency thresholds 0.95 and 0.5. These pointwise Monte Carlo intervals do not provide simultaneous coverage. Exact Clopper–Pearson intervals and all subgroup scopes remain in Online Resource 2, `r3/analysis/results/rev3_ladder_summary.json`.

Pair-free O detection and higher-weight audit.

The split-alpha Pearson histogram test uses fixed bit string bins, without pair selection, and detects all 50 O subjects (31 groups) in exact laws and stored shots. It was designed after archive and CE-result exposure; its fixed family is not a prospective discovery claim. The pair (q_2, q_3) explains the correlation, without being required by the histogram test. The higher-weight audit computed all 16,384 coefficients per fixed input and used no first-gap stop; fallback count is 0. Its minimum weight is conditional on that input, distinct from the negative check over all 128 basis inputs, 4 slots and 210 Pauli statistics of weight at most two. The exhaustive space, ordering and tie rule remain in Online Resource 2, `r3/analysis/results/rev3_audit_Q4.csv`.

Table S55 Source-tool outcome and original-binary dependency audit

Evidence	Interpretation
Muskit source failures	1,129 exact member/SHA256 links; executor E, C and O 204 (18.1%); slot-consistent E, C and O 374 (33.1%). Reported failure includes `P-Value nan`. 7 unlinked subjects omitted, without near-name substitution.
QuCheck source ratio	Not derivable: partial notebook rows do not supply the full source outcome and generated-instance crosswalk.
CPU reconstruction	Stored QASM/source census, contract-relative distinguishing inputs, stored-count reduction and CPU probability sampling do not require the original QuCheck GPU binary.
Native records	Stored native outcomes can be audited. Their missing assertion-to-circuit map prevents independent reconstruction of native p-values; a fresh native execution is a different claim.
Original execution	Generating original QuCheck shot records anew depends on the unpinned binary/interpreter environment. Historical runtime is a recorded observation, not a fresh-cost guarantee.
Missing observations	Unavailable historical resampling records cannot be reconstructed by obtaining a binary.
Locator inventory	1,312 source locations from the current manuscript, Online Resource 1 and historical source tables; repeated statements are retained, and unused rows are marked. The binding file records exact audited source hashes.

The cross-tab is exploratory, conditional on supplied-wrapper classes, and is neither a source-tool false-positive rate nor a revised mutation score. Exact IDs, hashes, scope and reconstruction limits: Online Resource 2, `r3/analysis/results/rev3_audit_Q5.json`, Online Resource 2, `r3/current_inventory/claims.csv`, Online Resource 2, `r3/current_inventory/Q6.json` and Online Resource 2, `r3/current_inventory/binding.json`.

Table S56 Every fixed-wrapper C subject and its recorded native outcome

Subject	ID prefix	Qubits	Native	Consequence
Deutsch–Jozsa m2	`5540ccc9`	3	non-detection	Native range unproved
Deutsch–Jozsa m16	`663b1d13`	3	non-detection	Native range unproved
Deutsch–Jozsa m12	`953255ba`	3	non-detection	Native range unproved
Grover m14	`25a82a39`	3	non-detection	Native range unproved
Grover m10	`a1a34288`	3	non-detection	Native range unproved
Grover m8	`b1bfc15e`	3	detected	Native counterexample
Teleportation m8	`c96ba53f`	3	non-detection	Native range unproved
Superdense coding m19	`34b06e6d`	2	non-detection	Native range unproved
Superdense coding m14	`f2215541`	2	non-detection	Native range unproved

Deutsch–Jozsa and Grover use fixed 3-qubit builds, teleportation 3 and superdense coding 2. Native Grover m8 separates outside that fixed instantiation. Other non-detections do not prove C over native ranges. Full generator ranges, seeds, source digests, count dimensions, properties and protocol hashes: Online Resource 2, `r3/analysis/results/rev3_audit_native_C.csv`.

S9 Details of the inherited-suite audit

This audit records dependence, equal-budget suite comparisons and the observed execution costs.

S9.1 Dependence and multiplicity

Every alternative detection beyond QuCheck's T_M lies in Bell, which contributes twice as many subjects as each of the other two lineages. Without Bell, both inherited suite and union detect 34 subjects; no suite exceeds this count (Table S12). Equal lineage weights give detected shares of 56.67% for T_M, 55.82% for the uniform mean and 57.78% for the union, including silent subjects. These shares weight lineages equally; raw shares weight subjects equally.

Grouping identical ordered per-slot QASM build-digest sequences within families leaves 117 subjects, with baseline, inherited-suite and union counts 49, 67, 68, uniform mean 65.24 and tail fraction 0.3091. These artefact groups differ from unitary behaviour groups (Table S26); neither grouping establishes independent faults.

S9.2 Calls, recorded time and census cost

Muskit's 165 baseline-preserving suites each use 4,544 target calls, but accumulated backend times span 20.368–22.199 s, a 9.0% spread relative to the minimum. The median is 21.619 s, while T_M takes 21.460 s, ranking 49 of 165 from shortest to longest (Table S28). These single-run sums exclude controls, construction, transport, scheduling and repeated-run effects; QuCheck worker times have a broader scope (Table S9).

The census used 22.21 s and 401,912 KiB peak resident memory for 5,024 builds of at most 7 qubits, including input search, adapter checks and grouping. Including control checks and output writing gives 26.22 s and 584,648 KiB; these are single-run observations. Dense double-precision complex storage grows as 16×4^n bytes per unitary, and equal target calls do not imply equal end-to-end budgets (Table S28).